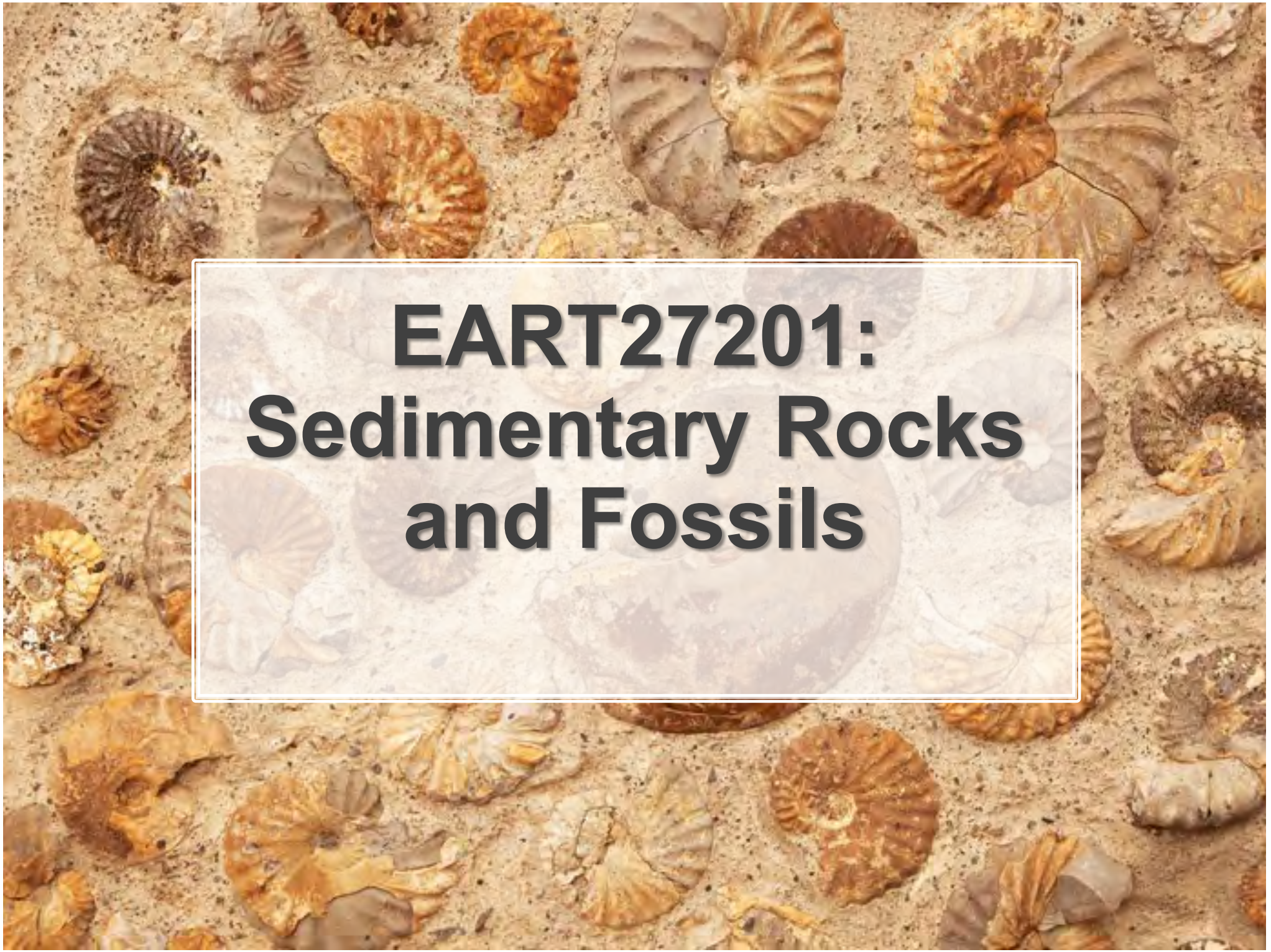

EART27201:
Sedimentary Rocks
and Fossils

Lecture outline

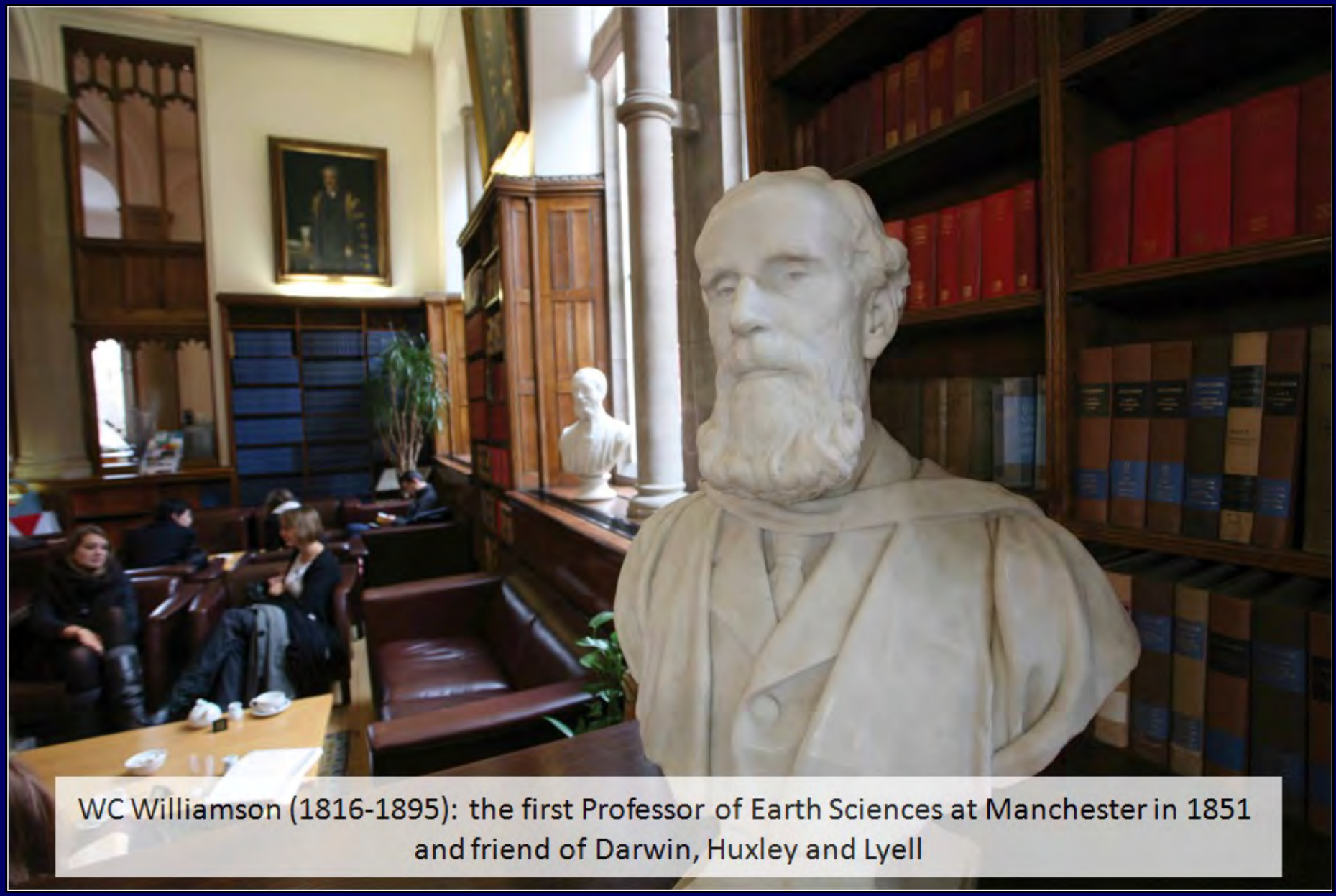
An introduction to this unit

- Why study sedimentary rocks and fossils?
- The aims of the unit
- Structure and delivery of the unit, including assessment
- Course texts

Followed by…

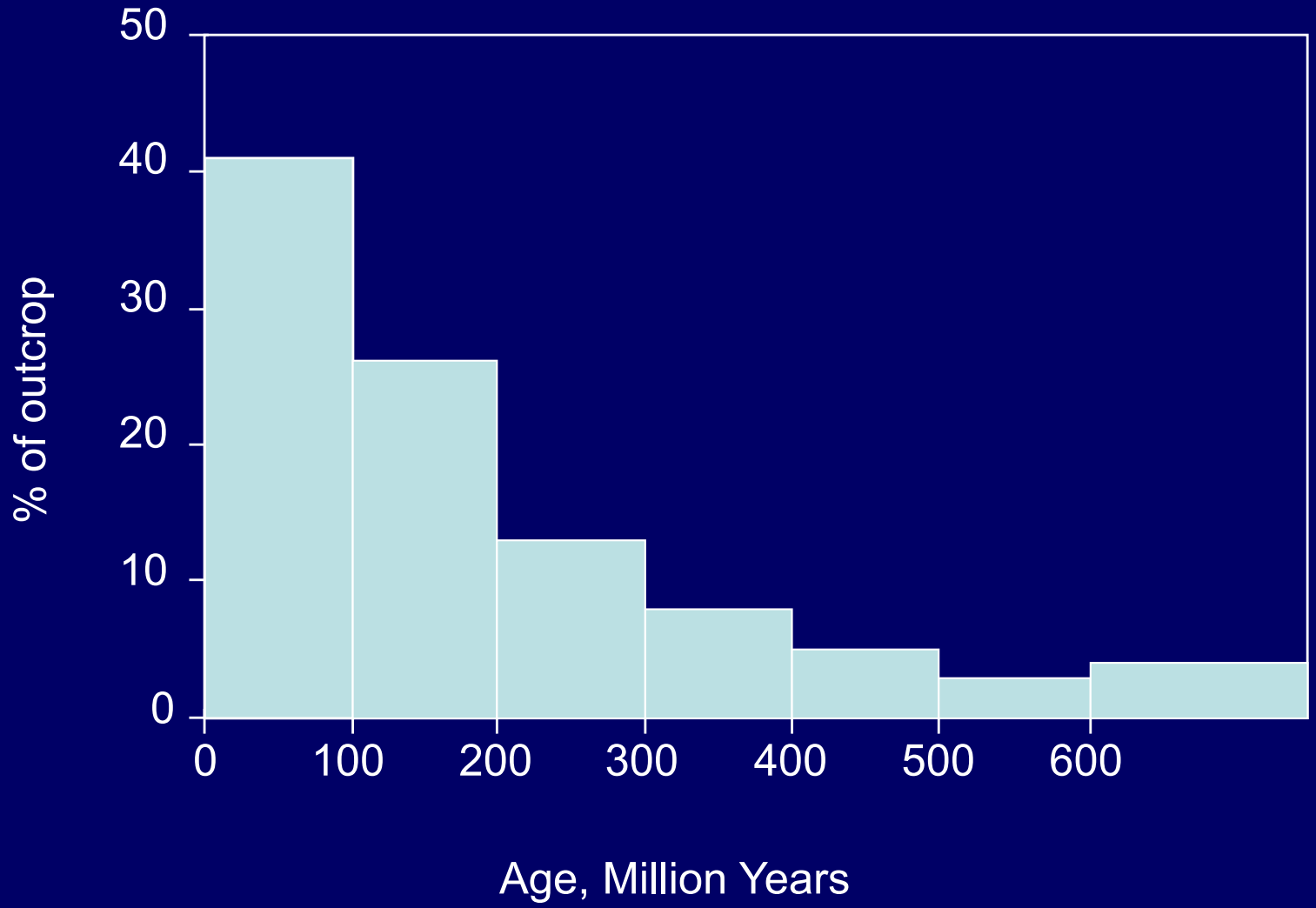
- Class 1: Nature, description and classification of sediments and sedimentary rocks

Studying palaeontology & sedimentology at Manchester



WC Williamson (1816-1895): the first Professor of Earth Sciences at Manchester in 1851 and friend of Darwin, Huxley and Lyell

Why study sedimentary rocks?



Sedimentary rocks cover ~75% of the Earth's surface

Why study sedimentary rocks?



Grand Canyon National Park, USA.

Sediments and sedimentary rocks contain:

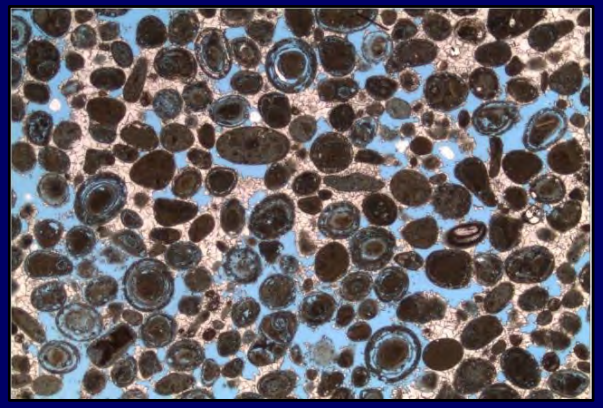
- all fossils and the entire record of Earth's geological history
  - evidence for environmental change and the Earth's response to that change, e.g., changing climate, growth and decline of supercontinents, past mountains and oceans, and the evolution of life

- source of economic minerals and materials
  - reservoirs of fossil fuels (oil and gas), aquafers for water
  - potential storage sites for CO_2, nuclear and other wastes
  - building and aggregate materials and many other essential raw materials e.g., coal, iron, salt, gypsum, source of cement, pottery clay, placer deposits

Unit aims

- provide you with the tools and language to be able to identify, describe and measure sedimentary rocks in terms of their material properties
- familiarize you with common groups of micro- and macro- fossils, and their applications to inferring the ages of rocks, and their environments of deposition
- equip you with the skills to use observations from the field and hand specimens to infer the transport, deposition and diagenetic histories of the sediment and subsequent rock







Structure, delivery and assessment





← **Dr. Merren Jones**, Sedimentologist and stratigrapher, with interests in physical experimentation, tectonics and sedimentation, and the ingestion of dust into aircraft engines.



→

Dr. Russell Garwood, Palaeobiology and evolution.

Structure, delivery and assessment

Blended approach

- Combined 3-hr lecture and hands on practical class
- G12/14 Williamson, 11am - 2pm, Thursday
- Class will start promptly at 11am
- If miss a week, practical materials available for catch up but lectures not recorded

Sedimentary Rocks (Merren)

3 Oct	The nature, description and classification of sediments
10 Oct	Physics of sedimentation
17 Oct	Unidirectional flows and formation of bedforms
24 Oct	Siliciclastic rocks: sedimentary structures and stratification
31 Oct	Carbonate rocks: carbonate deposition, limestone classification, and porosity
7 Nov	Introduction to stratigraphy, facies and facies models

Fossils (Russell)

14 Nov	Macrofossils: Introduction and corals
21 Nov	Macrofossils: Molluscs and brachiopods
28 Nov	Macrofossils: Trilobites and echinoderm
5 Dec	Macrofossils: Graptolites and conclusion
12 Dec	Microfossils

Structure, delivery and assessment

Assessment comprises……

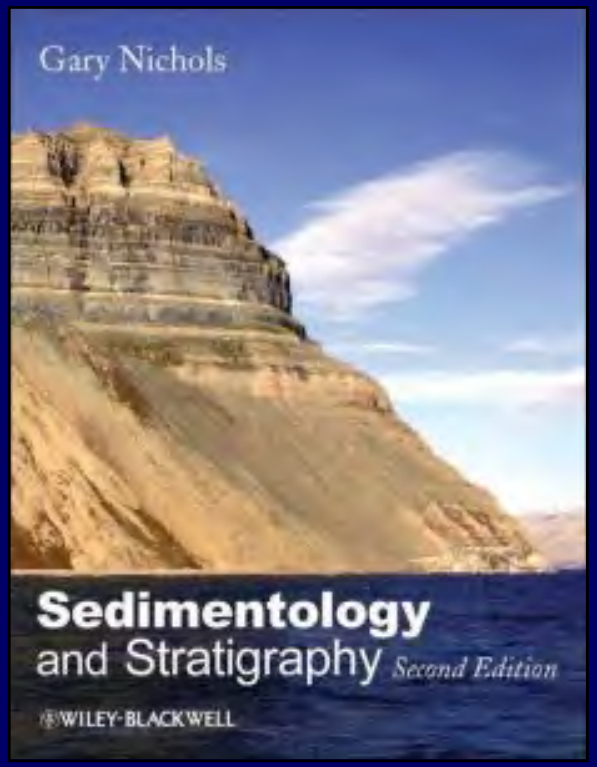
Course work (50%)

- 10 weekly, on-line, 10-minute tests
- Must complete on own, closed-book on Thursday between 9am and 6pm: first test 9 October
- Covers material from previous Friday
- Feedback in class on following Friday

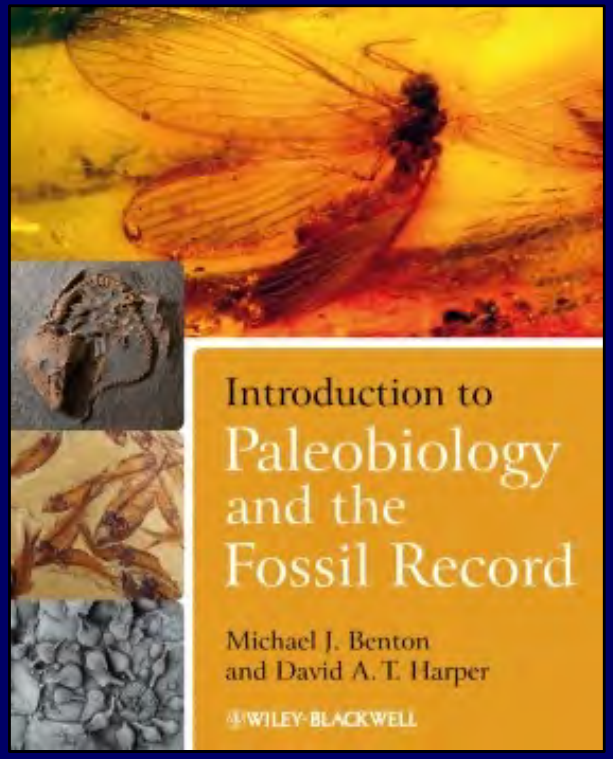
Exam (50%)

- January 2026 in exam period

Course texts

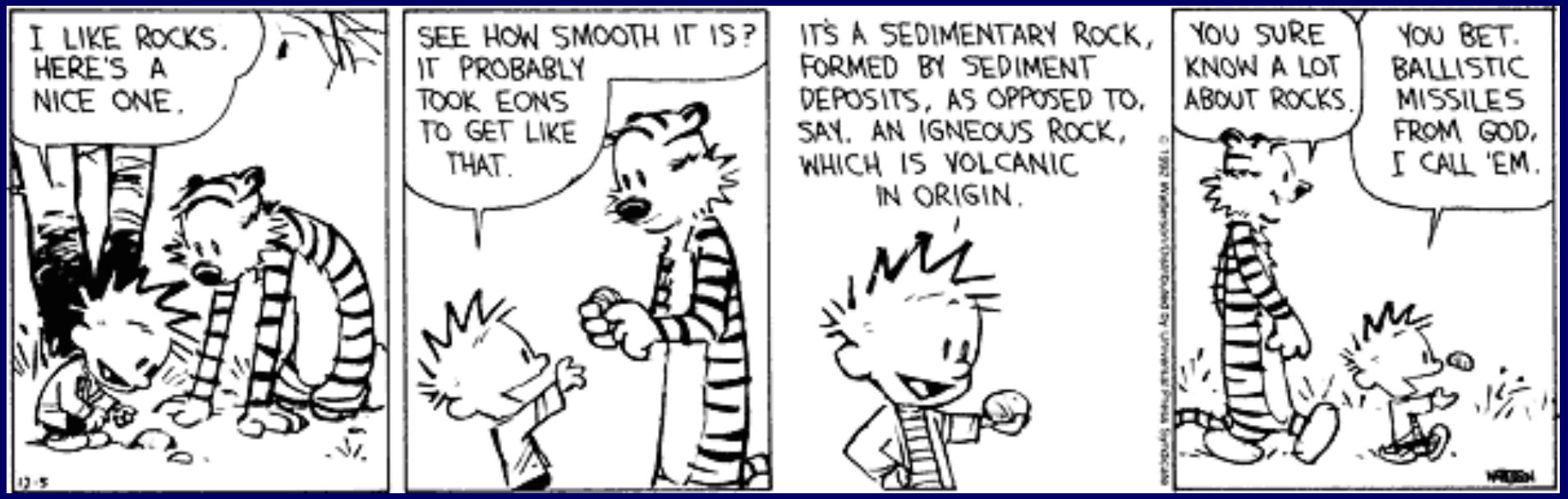


- Nichols, G. (2009) Sedimentology and Stratigraphy, 2nd Edition, Wiley-Blackwell Publishing.
- Available from the University Library as an e-book.



- Benton, M.L., and Harper, D.A. (2009) Introduction to Paleobiology and the Fossil Record, Wiley-Blackwell Publishing.
- Available from the University Library as an e-book.

The Nature, Description and Classification of sediments



Calvin and Hobbes

Lecture overview

- Kinds of sedimentary rocks
- Sediment size and shape
- Other things about the texture of detrital sediments
- The origin of particles
- Sedimentary minerals
- Classification of siliciclastic rocks

Kinds of Sedimentary Rocks

Clastic (detrital): Physical Processes

allochthonous

Terrigenous: siliciclastic rocks (consist of fragments of pre-existing rocks, transported and deposited by physical processes e.g. cg, ss, ms)

Penecontemporaneous volcanic activity: volcaniclastics (consist of lava and rock fragments e.g. tuffs, ignimbrites etc.)

Non-clastic: Chemical and Biological Processes

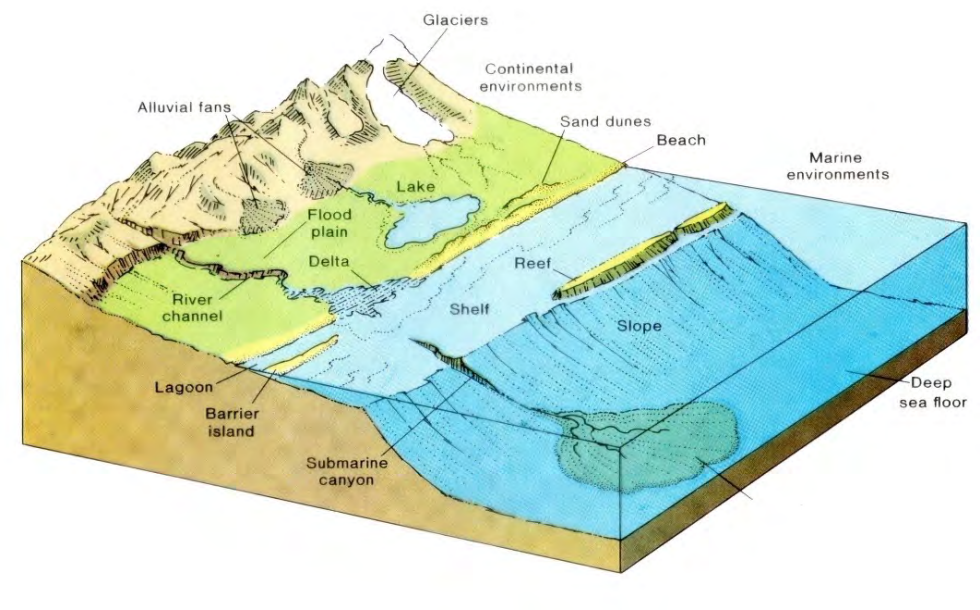
autochthonous

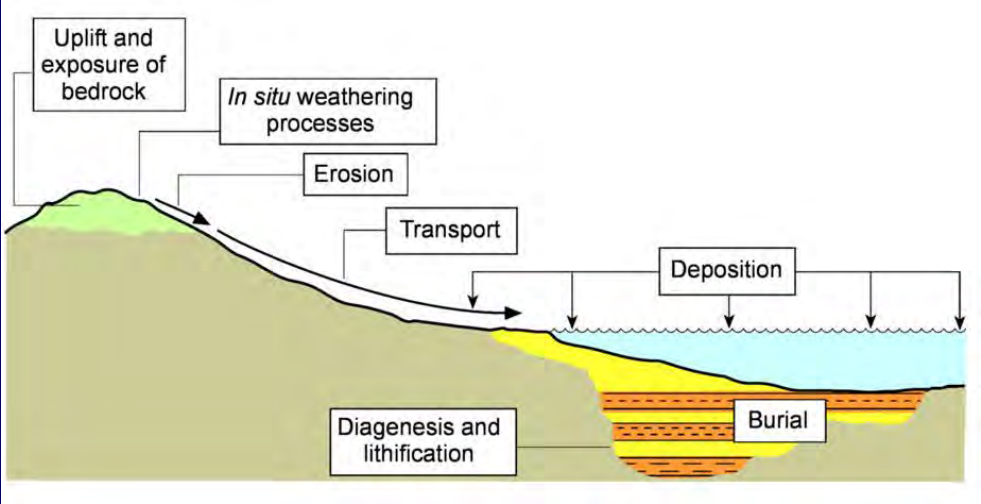
Biogenic, biochemical and organic processes: e.g. limestone, coal, oil shale and chert (oozes)

Chemical processes: evaporites, flints and ironstones

How terrigenous clastic sedimentary rocks are made

The ultimate recycled product!





Clastic sediment is derived directly or indirectly from the weathering and erosion of subaerially exposed pre-existing rock

- Made available via uplift and exposure
- Tectonic and volcanic processes

Study of sedimentary rocks involve all stages in their evolution

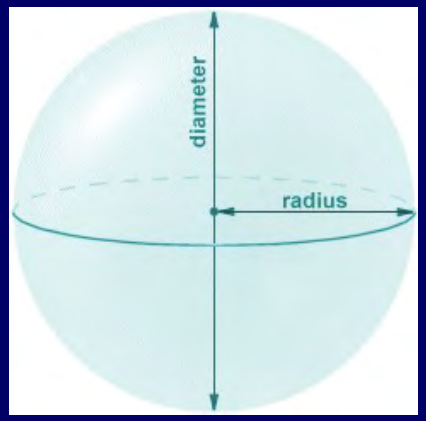
Description of clastic sediment must describe particles in terms of both individual (size, shape, composition) and bulk properties (are grain size distribution, texture, sedimentary structures, porosity and permeability)

Sediment (*a.k.a.* particle, clast, grain) size

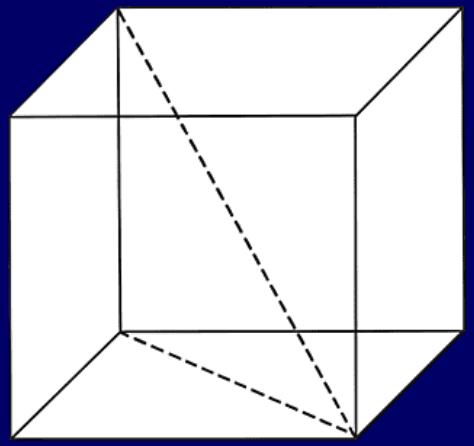
Why measure size?

Fundamental characteristic: basis for comparing sediments, indicative of transport and depositional conditions, related to other properties

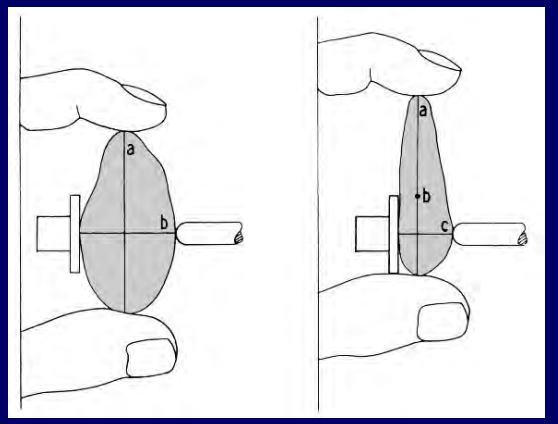
How to measure size?



Sphere



Cube



Triaxial ellipsoid

Problem: *sedimentary particles are almost always irregular in shape*

What do we mean by size of a representative particle?

Could characterize the size of the particle by its nominal diameter, the size of the sphere with volume equal to that of the given particle

- nice concept, but hard to put into practice

There is no single unique dimension we can measure readily that would suffice to characterize the size of the particle

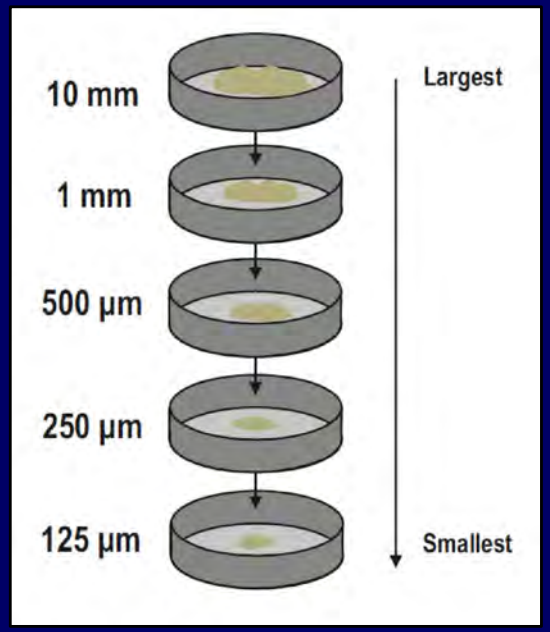
How to define and measure particle size

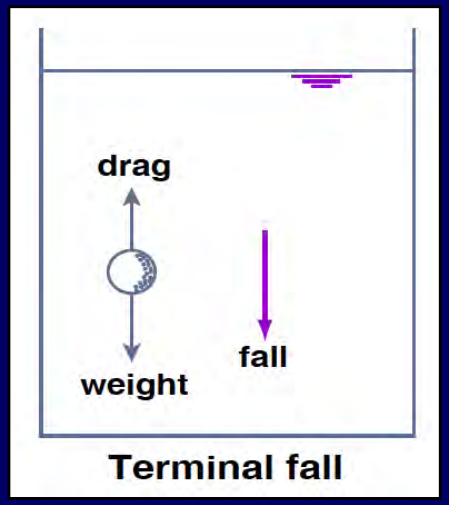
Two approaches:

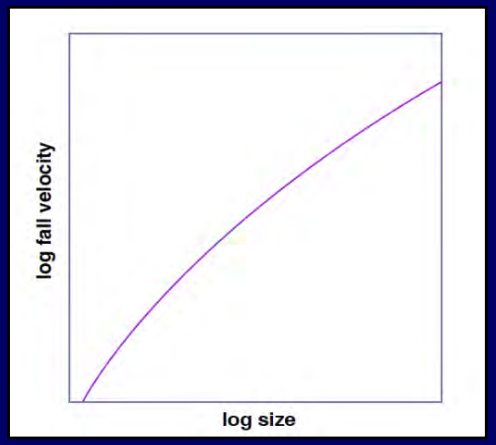
How we measure size defines what we mean by size
e.g., **sieving**

- standard for coarser-grained sediment (above several 10s of microns)
- sieves choose the largest dimension in the least cross-sectional area







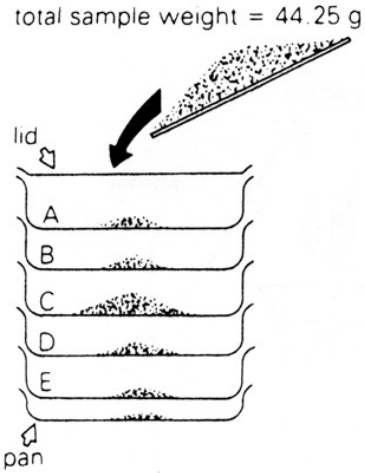


Relate size to some other more easily defined property
e.g., **settling**

- standard for finer-grained sediment (silt and clay, but also sand)
- larger particles fall through water faster than smaller particles
- assumes particles are spheres, with density of quartz, settling in water at a standard temperature

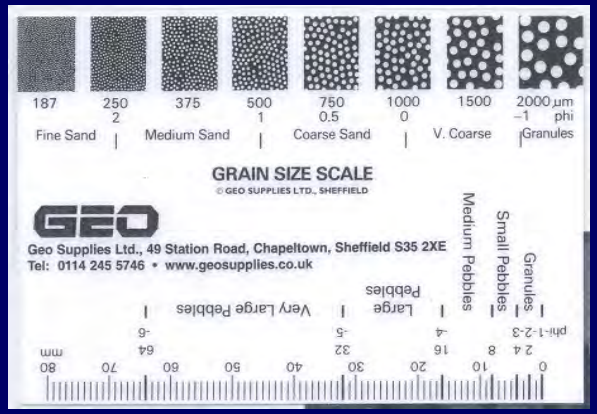
In the practical today….

Part A: Grain size analysis of unconsolidated sediment



Sieve	Screen size (mm)	Weight of sieve (g)
A	1 0	5.31
B	0 5	8.70
C	0 25	12 59
D	0 125	9 13
E	0 0625	6 03
pan	<0 0625	2 49

Part C: Grain size analysis of sedimentary rocks in hand specimen



Grain size comparator card



Binocular microscope

Udden-Wentworth grade scale

Length (mm)			Grain size descriptor	Sediment/ rock name
	ϕ		block	mega-conglomerate
4096	−12	vc	boulder	gravel conglomerate
2048	−11	c		
1024	−10	m		
512	−9	f		
256	−8	c	cobble	
128	−7	f		
64	−6	vc	pebble	
32	−5	c		
16	−4	m		
8	−3	f		
4	−2		granule	
2	−1	vc	sand	sand sandstone
1	0	c		
0.50	1	m		
0.25	2	f		
0.125	3	vf		
0.063	4	c	silt	silt siltstone
0.031	5	m		
0.015	6	f		
0.008	7	vf		
0.004	8		clay	clay claystone

Fundamental characteristic of a terrigenous clastic rock is its grain size (~1 mm - >10 m)

Wentworth

Fundamental division into Gravel, Sand and Mud with boundaries at 2mm and 0.063mm

If lithified:

Gravels become conglomerates/breccias or rudites

Sands become sandstones or arenites

Muds become mudstones or argillites

Udden-Wentworth grade scale

Grains easily visible to the naked eye and measurable





Clasts visible to the naked eye (finer sizes may require aid of hand lens)

Clasts not visible to the naked eye



Udden-Wentworth grade scale

Length (mm)			Grain size descriptor	Sediment/ rock name
	ϕ		block	mega-conglomerate
4096	-12			
2048	-11	vc	boulder	gravel conglomerate
1024	-10	c		
512	-9	m		
256	-8	f		
128	-7	c	cobble	
64	-6	f		
32	-5	vc	pebble	
16	-4	c		
8	-3	m		
4	-2	f		
2	-1		granule	
1	0	vc	sand	sand sandstone
0.50	1	c		
0.25	2	m		
0.125	3	f		
0.063	4	vf		
0.031	5	c	silt	silt siltstone
0.015	6	m		
0.008	7	f		
0.004	8	vf		
			clay	clay claystone

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Sands become sandstones or arenites

Muds become mudstones or argillites

Udden

Further subdivision uses powers of 2

Using mm as the units, scale is geometric (i.e., 1, 2, 4, 8, 16)

When doing statistics on grain sizes a linear (arithmetic) scale is used
(i.e., 1, 2, 3, 4 etc.)

Udden-Wentworth grade scale: ϕ scale

Length (mm)	ϕ		Grain size descriptor	Sediment/ rock name
			block	mega-conglomerate
4096	-12	vc	boulder	gravel conglomerate
2048	-11	c		
1024	-10	m		
512	-9	f		
256	-8	c	cobble	
128	-7	f		
64	-6	vc	pebble	
32	-5	c		
16	-4	m		
8	-3	f		
4	-2		granule	
2	-1	vc	sand	sand sandstone
1	0	c		
0.50	1	m		
0.25	2	f		
0.125	3	vf		
0.063	4	c	silt	silt siltstone
0.031	5	m		
0.015	6	f		
0.008	7	vf		
0.004	8		clay	clay claystone

When doing statistics on grain sizes a linear (arithmetic) scale is used
(i.e., 1, 2, 3, 4 etc.)

A logarithmic function is needed to convert a scale in powers of 2 to a linear scale.

$\phi = - \log_2(\text{diameter in mm})$

This is called the ϕ (Phi) scale.

In practical terms, this means coarse grain sizes have high negative values and fine grain sizes positive values.

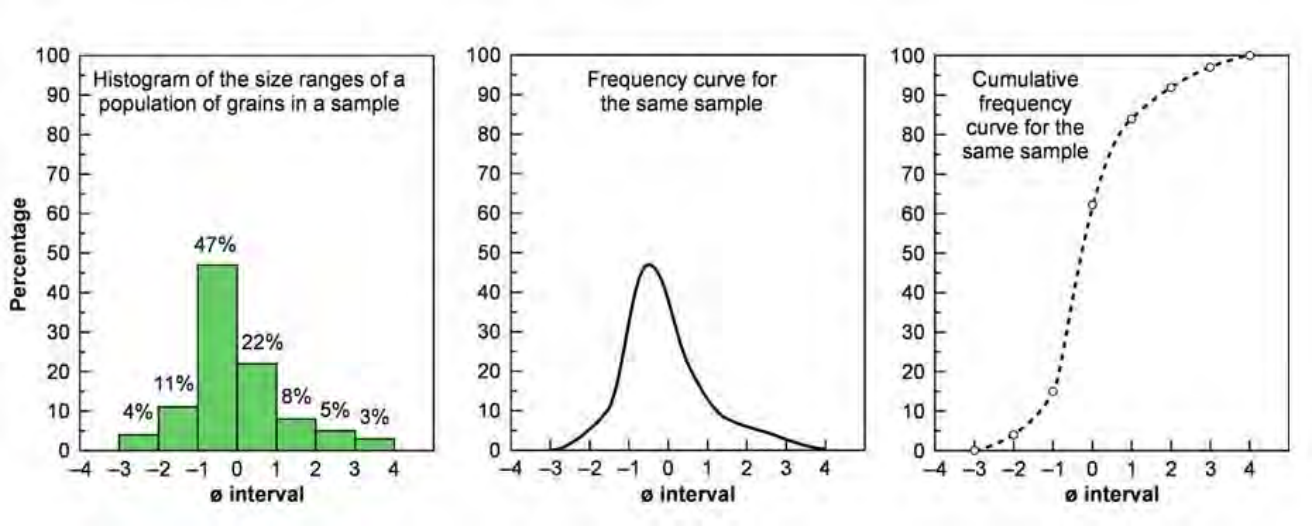
In describing sediments in lab and field you will use mm. You will only use ϕ when doing statistics.

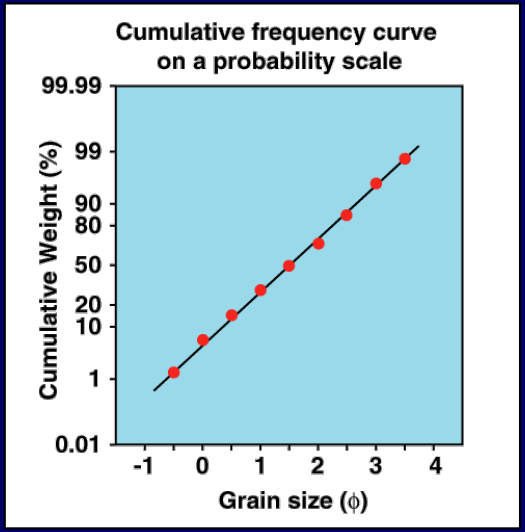
millimeters	4	2	1	1/2	1/4	1/8	1/16	1/32
ϕ	-2	-1	0	1	2	3	4	5

A conversion table between millimeters and phi units

Statistical analysis of particle size

Graphic presentation of data





Key variables

Can be calculated using Method of Moments or using percentile measures off a cumulative frequency graph

Note: always plot the cumulative curve on graph paper with a probability ordinate (vertical axis) when measuring percentile values

Average size

- Mean size
- Median
- Mode

Measure of dispersion (*a.k.a.* sorting)

- Standard deviation

Degree of asymmetry of the distribution

- Skewness

Practical Exercise 1

Complete Part A. Grain Size Analysis

Classification of sand-gravel-mud mixtures

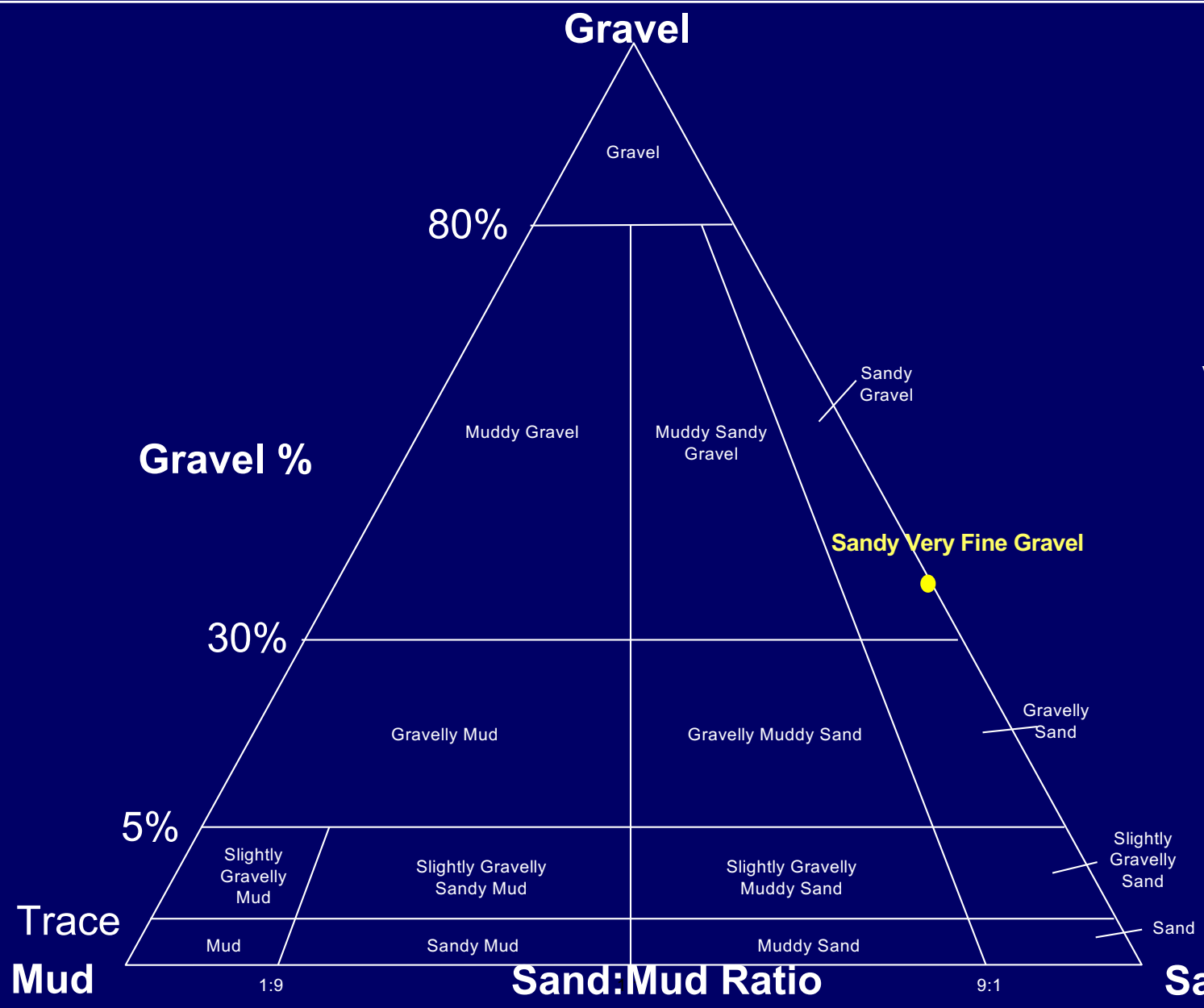
A simple classification of terrigenous clastic rocks and sediment is based on the predominant grain size of the material:

Grain Size[1] (mm)	Sediment name	Rock Name	Adjectives
> 2	Gravel	Conglomerate	Cobble, pebble, well sorted, etc.
0.0625-2	Sand	Sandstone	Coarse, medium, well sorted, etc.
< 0.0625	Mud	Mudstone	Silt or clay

[1]In Part A, for the purposes of this general classification we assigned the rock or sediment name based on the mean or median grain size (in hand specimen, the size range in which there are >50% of the grains.

e.g., Sample 1, mean grain size -0.72ϕ, very coarse sand

More detailed classification schemes will limit terms on the basis of different proportions of sediment within a given grain size.



Gravel:	37.7%
Sand:	62.3%
Mud:	0.0%

Very Coarse	0.0%
Coarse Gravel:	0.0%
Medium Gravel:	6.3%
Fine Gravel:	11.3%
Very Fine Gravel:	20.1%
Very Coarse Sand:	32.1%
Coarse Sand:	24.7%
Medium Sand:	4.4%
Fine Sand:	0.8%
Very Fine Sand:	0.3%
Very Coarse Silt:	0.0%
Coarse Silt:	0.0%
Medium Silt:	0.0%
Fine Silt:	0.0%
Very Fine Silt:	0.0%
Clay:	0.0%

(after Udden-Wentworth and Blair & McPherson, 1999)

Particle shape: Sphericity

Textural aspects of particles

Shape (sensu stricto): the whole-particle aspect of shape; this is usually viewed with reference to a sphere, so it's termed *sphericity*

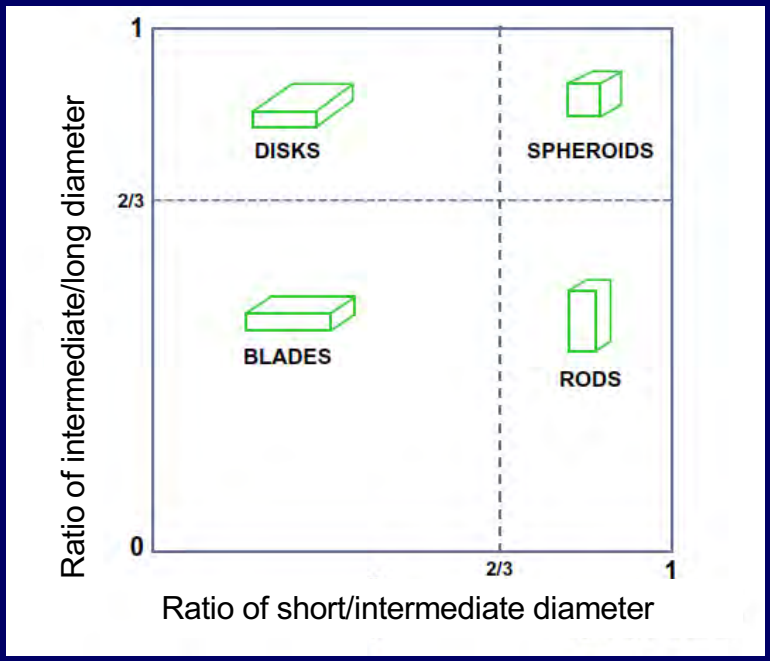
Roundness (angularity): the local aspect of shape, involving the sharpness, or lack thereof, of edges and corners

Sphericity

Definition: the ratio of the nominal diameter of the particle to the diameter of the circumscribed sphere



A sphere has a sphericity of one

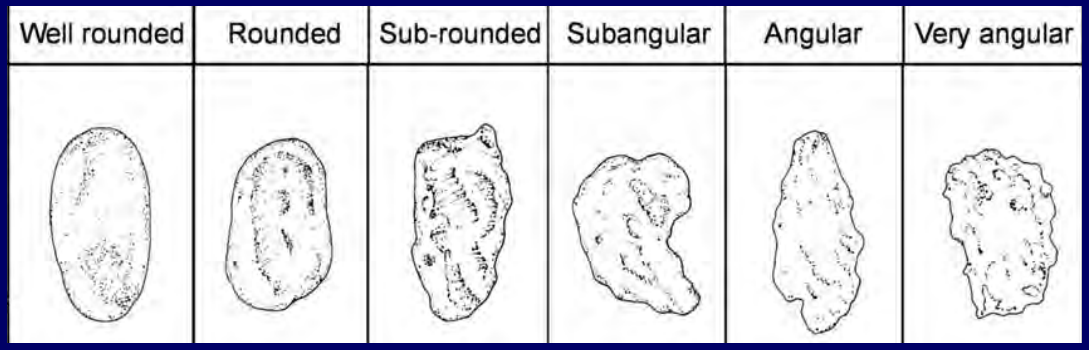


The Zingg diagram, to describe the shape of sediment particles that can be approximated by triaxial ellipsoids

Particle shape: Roundness/Angularity

Roundness: the local aspect of shape, involving the sharpness, or lack thereof, of edges and corners

Definition: the ratio of average radius of curvature of edges and corners to the radius of the largest inscribed sphere



angular: little evidence of wear; edges and corners sharp

subangular: some evidence of wear; edges and corners rounded to some extent; original faces only slightly modified

subrounded: considerable wear; edges and corners rounded; area or original faces reduced

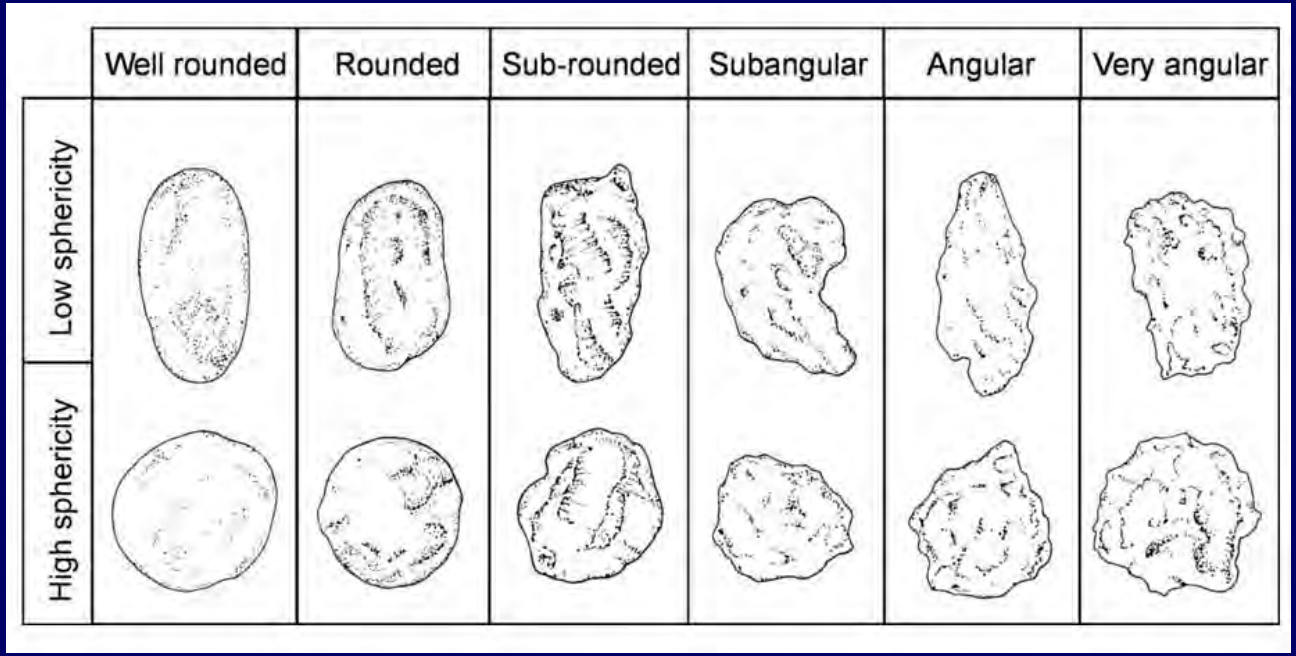
rounded: original edges and corners smoothed to broad curves; original faces almost gone; original shape still apparent

well rounded: no original edges, corners or faces; original shape only suggested by present shape

Qualitative descriptions of shape grades

In the practical today….

Part C: Sandstone hand specimens



Note: the sharpness of edges and corners of a rock or mineral particle, and is almost entirely independent of shape



Binocular microscope

Things about sediment

What controls the size of particles in sediment?

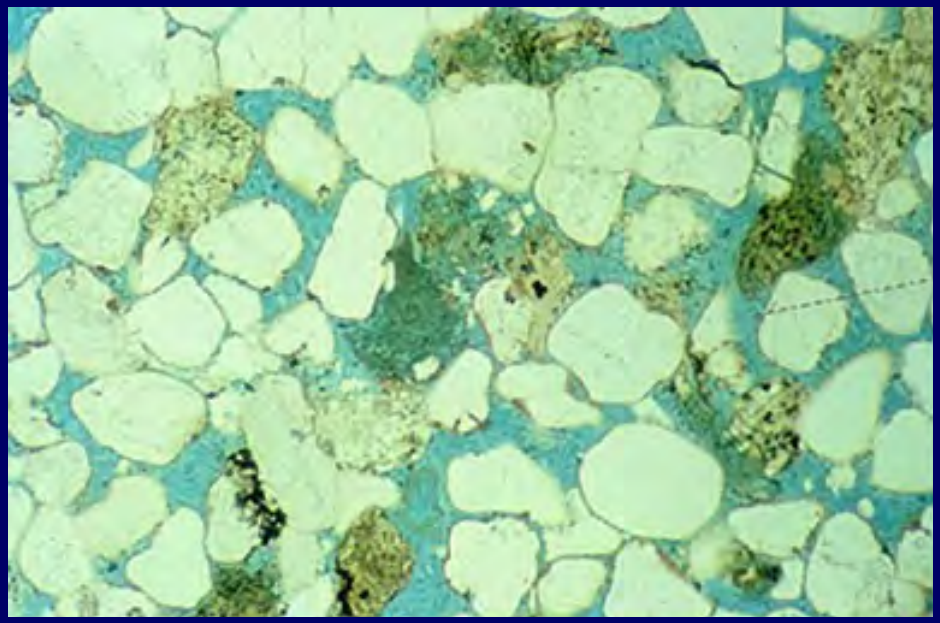
- the size of material available, and
- what has happened during transportation and deposition (complicated)

What might shape and roundness of a particle depend on?

- original shape and roundness upon production (crystallization; fracture)
- nature of weathering
- inherent nature of material (crystal habit; mineral/rock cleavage; abrasional anisotropy; fracture anisotropy)
- transport processes: distance, mode, medium (abrasion; corrosion; breakage)

Other aspects of texture: packing (*a.k.a.* fabric)

Packing or fabric: the mutual arrangement of grains in the deposit



Width of field of view approx. 3mm

Thin section of a sandstone showing intergranular porosity

Pores are filled with blue-dye impregnated resin

Porosity: amount of void space, expressed as a volume %

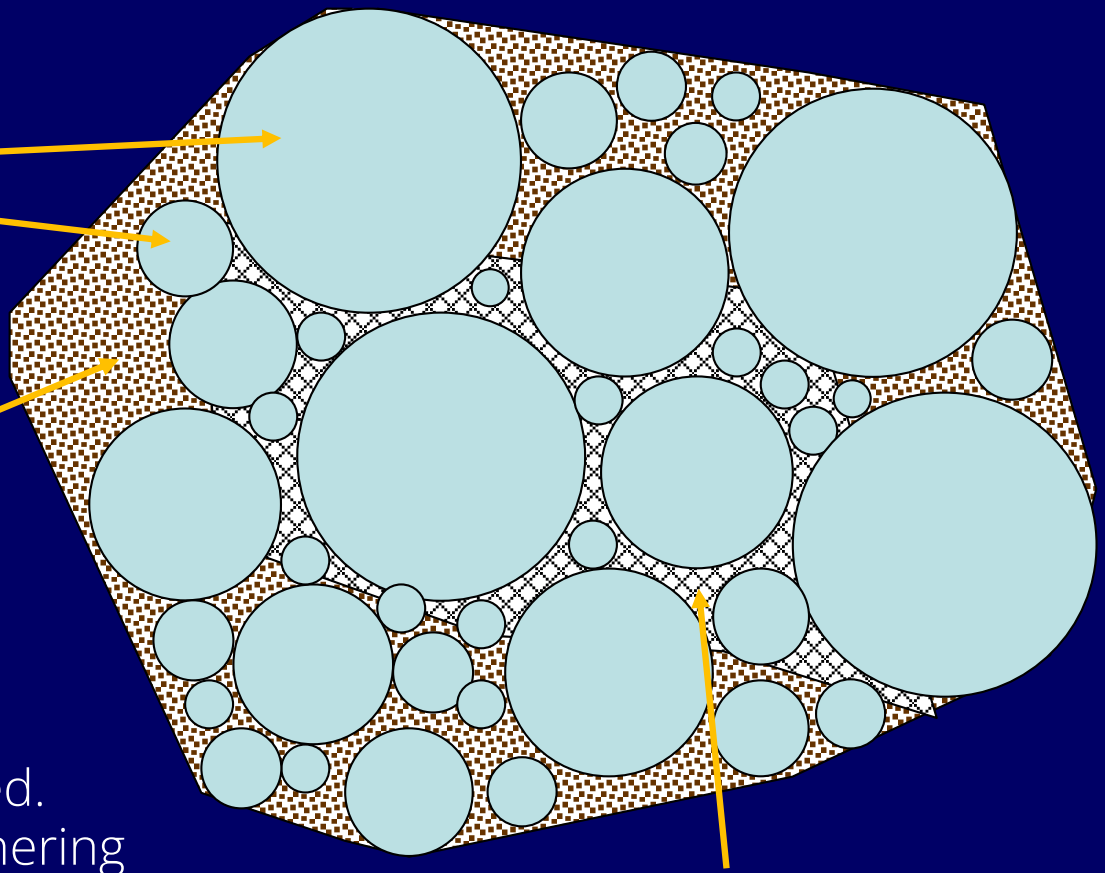
- may be filled with air, water or, occasionally, hydrocarbon
- some porosity is primary – formed at the time of deposition
- other porosity is secondary – it has appeared in the rock as a result of some post-depositional process

Permeability: describes the ease with which fluids can be forced through the porous sediment or rock under the influence of a gradient in fluid pressure

How to look at the geometry of a sediment or rock

In hand specimen or thin-section the standard way is:

Framework grains
a.k.a. clasts or particles
residual components of weathering i.e. quartz, feldspar and lithic fragments



Matrix
'fines'. Particulate material but altered. Typically products of chemical weathering such as clay minerals and silica

Cement
formed after deposition by precipitation into spaces

Compositional components of sedimentary rocks

Terrigenous or siliciclastic rocks

- quartz
- feldspars
- rock (*a.k.a.* lithic) fragments
- micas
- heavy minerals
- clay minerals

Carbonate rocks

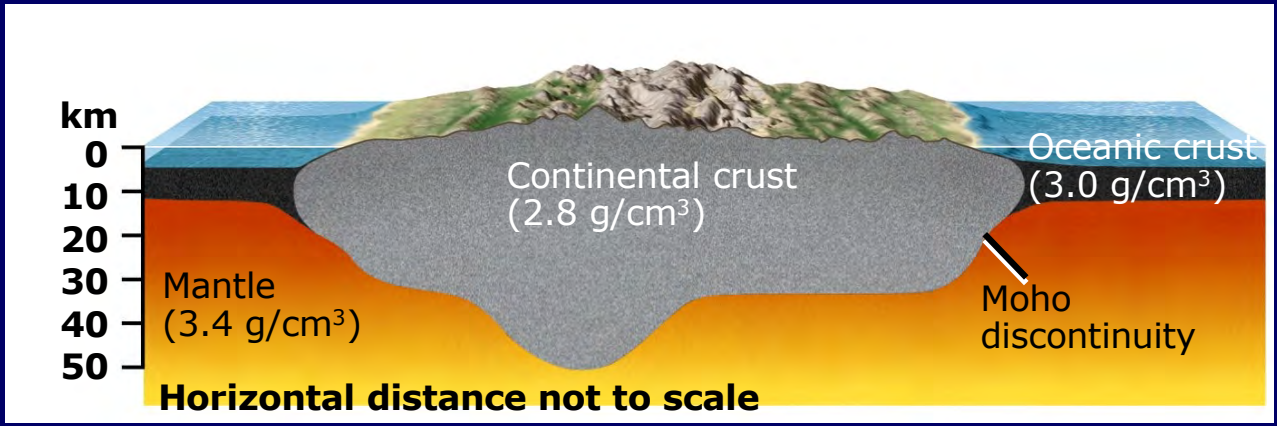
- calcite + aragonite
- dolomite

General key points to keep in mind

- *Any mineral can be found in a siliciclastic sedimentary rock*, but *not many are common*
- There are *fewer important terrigenous sedimentary minerals* than there are important igneous and metamorphic minerals *Why?*
- *Most minerals in terrigenous sedimentary rocks are metamorphic* (but by volume igneous minerals are more abundant)
- The *same minerals don't have the same significance in a sedimentary rock as in igneous and metamorphic rocks*: *minerals in terrigenous sedimentary rocks are not necessarily in equilibrium with each other or their environment*
- The *same mineral can be present in a rock for both equilibrium and disequilibrium reasons* (i.e. the same mineral can be in both detrital and authigenic forms)

Origin of clastic sediment

Weathering of continental crust is primary source of terrigenous clastic material



Continental crust:

- composed of felsic rocks e.g. granites and gneiss
- bulk composition "intermediate" (Andesitic or granite-like)

Comparison between the composition of continental crust and terrigenous clastic sedimentary rocks

Mineralogy	Igneous and Metamorphic
Feldspar	60%
Quartz	25%
Ca,Mg/Fe silicates (incl micas)	15%

Discrepancy results from interaction of rocks with water and dissolved atmospheric gases during weathering

Major weathering processes

Chemical weathering (always mediated by water)

- → Hydrolysis
- → Hydration
- → Carbonation
- → Oxidation
- → Ion exchange
- → Chelation

Lead to

- → Formation of new minerals with different volumes
- → Dissolution of minerals



Chemical decomposition of granite

Physical weathering

- → Frost wedging (freeze-thaw)
- → Exfoliation (heat-cool, peeling)
- → Salt weathering
- → Action of plants
- → Stress release

Rocks are broken into smaller fragments, but without significant change in chemical or mineralogical composition



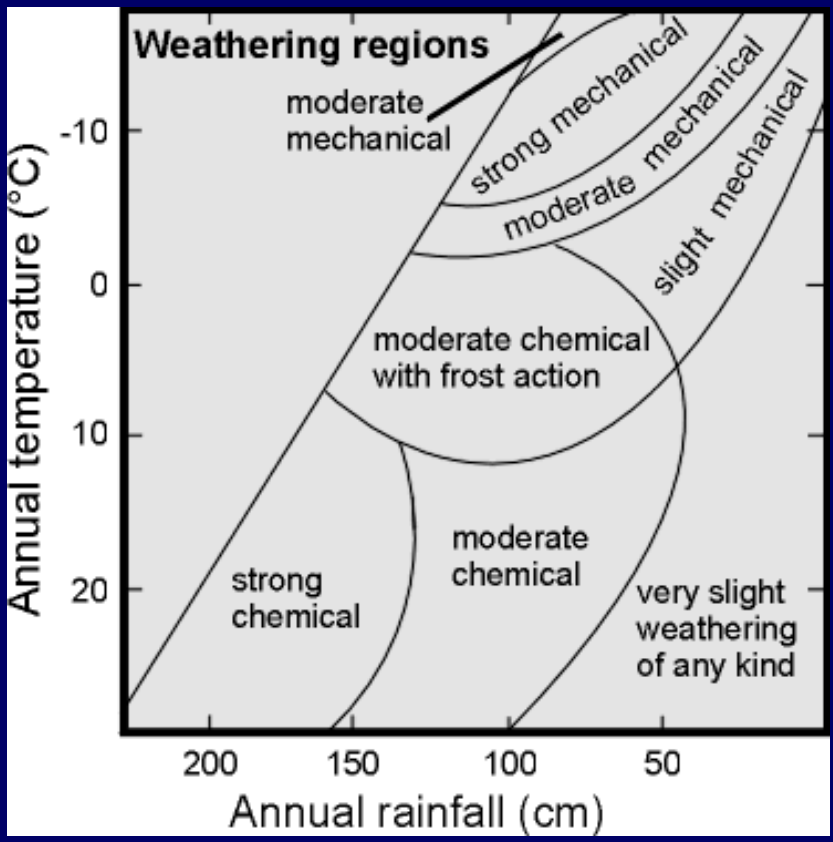
Fracturing due to freeze-thaw

Controls on weathering and erosion rates

Climate (temperature and rainfall) and relief of the source area are the main controls

Climate controls the *intensity* of weathering

- *Chemical reactions more rapid at higher temperatures*
- *Areas of high rainfall provide water needed for chemical weathering*
- Humid Tropics: chemical weathering is most rapid: intense leaching of both silicates and carbonate
- Temperate climates: chemical weathering less rapid (water in solid phase so can't mediate chemical reactions) but freeze/thaw becomes important particularly in upland areas, so mechanical weathering dominates
- Arid climates (hot or cold): weathering rates are low



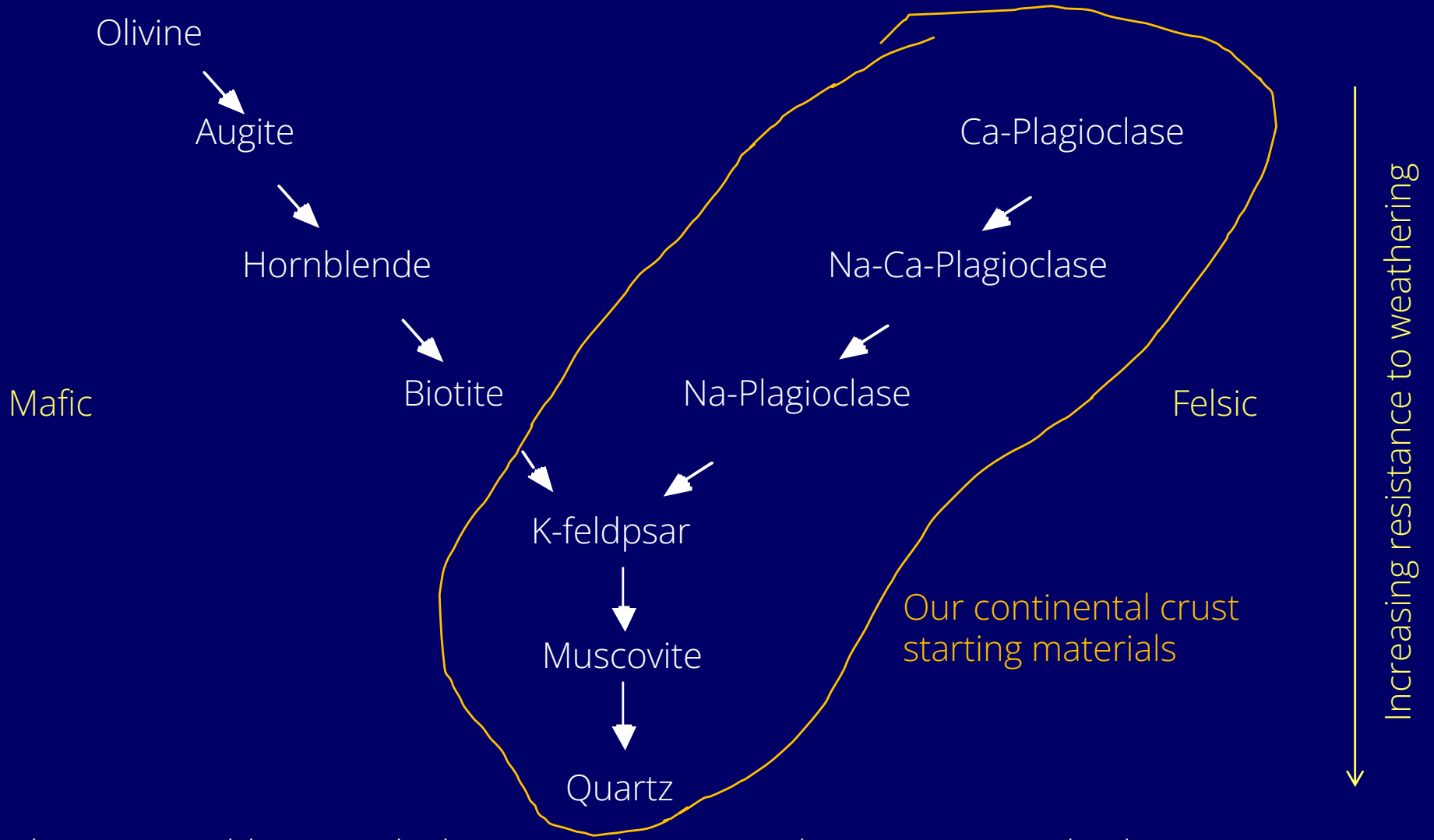
Relief controls the *time available* for weathering

- High relief = rapid erosion and short time for weathering
- Low relief = rates of erosion slow and longer time for weathering

Susceptability of minerals to weathering

Goldich Weathering Series

Stability series is reverse of order of crystallization of minerals from magma (Bowen's Reaction Series)



Note that most stable minerals during weathering are those precipitated at lowest temperatures.

Quartz (SiO_2)

Most common mineral in sediments in continental crust (av. 25%; can be 100% in sandstones but low % in fine-grained rocks and carbonates)



Main characteristics of quartz:

Effectively inert at the Earth's surface

Stable at surface T and P, and under all T and P conditions of sedimentary processes, as well as under almost all igneous and metamorphic conditions

Solubility in pure water is extremely small

→ Wont chemically weather

Hardness and lack of cleavage

→Highly resistant to physical weathering and transportation (at least in water)

As a result, sediment will have a similar % to source rocks

Provenance

- Majority derived from plutonic granitoid rocks, acid gneisses and schists
- Also, quartz bearing sedimentary rocks

Quartz is common to begin with, and is difficult to destroy.

Quartz (SiO_2)

Most *gravel-size* quartz comes from faults or mineral veins and is called vein quartz

Usually is milky white in appearance.

Sand-size quartz comes from crystalline source rocks e.g. granites, recycled quartz-rich sandstones or chert

In sandstone hand specimens, quartz is clear, glassy or slightly milky, but never dull white

Silt-size quartz grains come from fine-grained metamorphic rocks, preexisting fine-grained sediments, and chipping and spalling of larger grains





Quartz (SiO_2)

In thin-section:

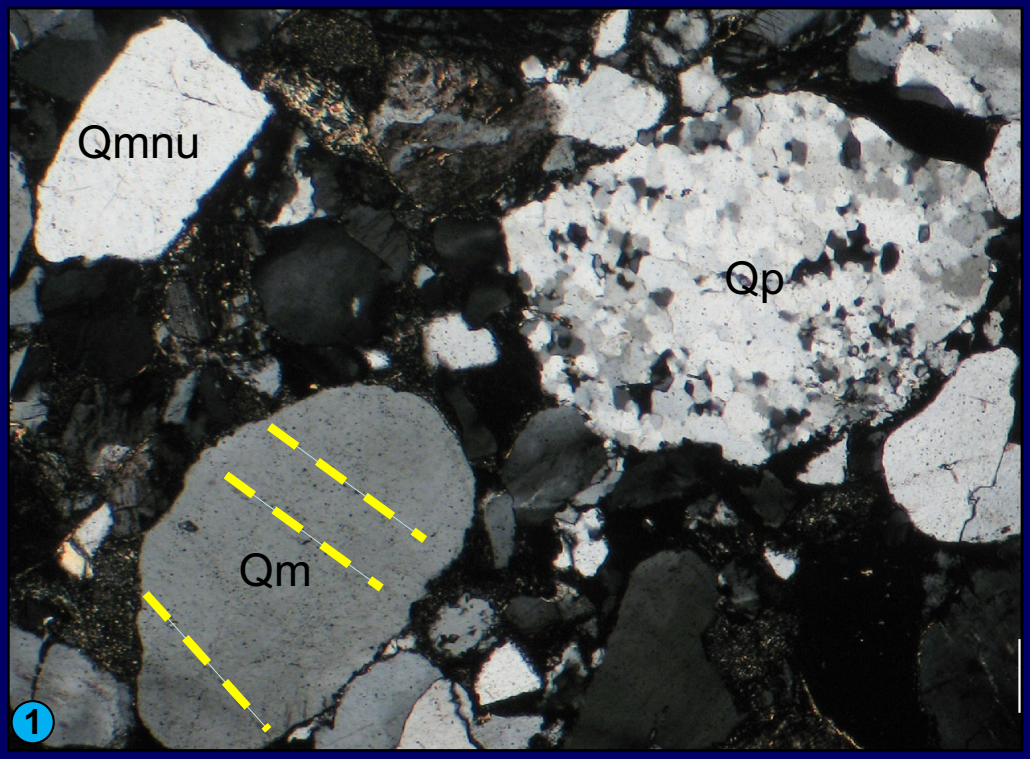
Monocrystalline grain (Qm)

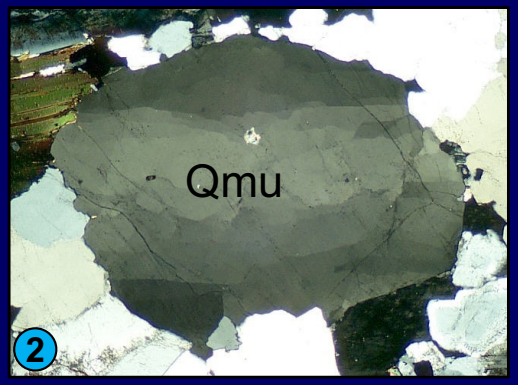
- single crystal
- basis of extinction:
  - uniform extinction under cross-polars (non-undulatory: Qmnu)
  - extinction sweeps across crystal as it is rotated over more than 5^0 (undulatory: Qmu) = reflection of strain in the crystal lattice

- presence of inclusions
  - vacuoles filled with fluid
  - minute crystals of other minerals

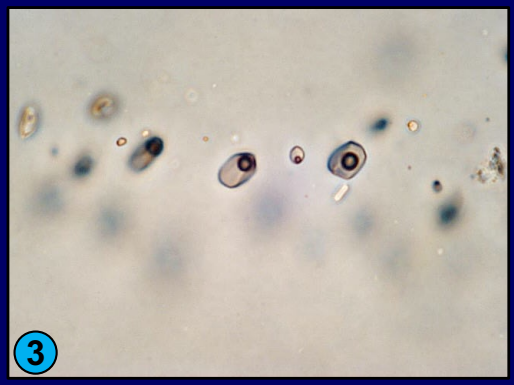
Polycrystalline grain (Qp)

- two or more crystals
- nature of boundary (straight, sutured, irregular)





Strained quartz crystal, XPL, 20x



Fluid containing (probably) water and a vapour bubble. PPL 1000x

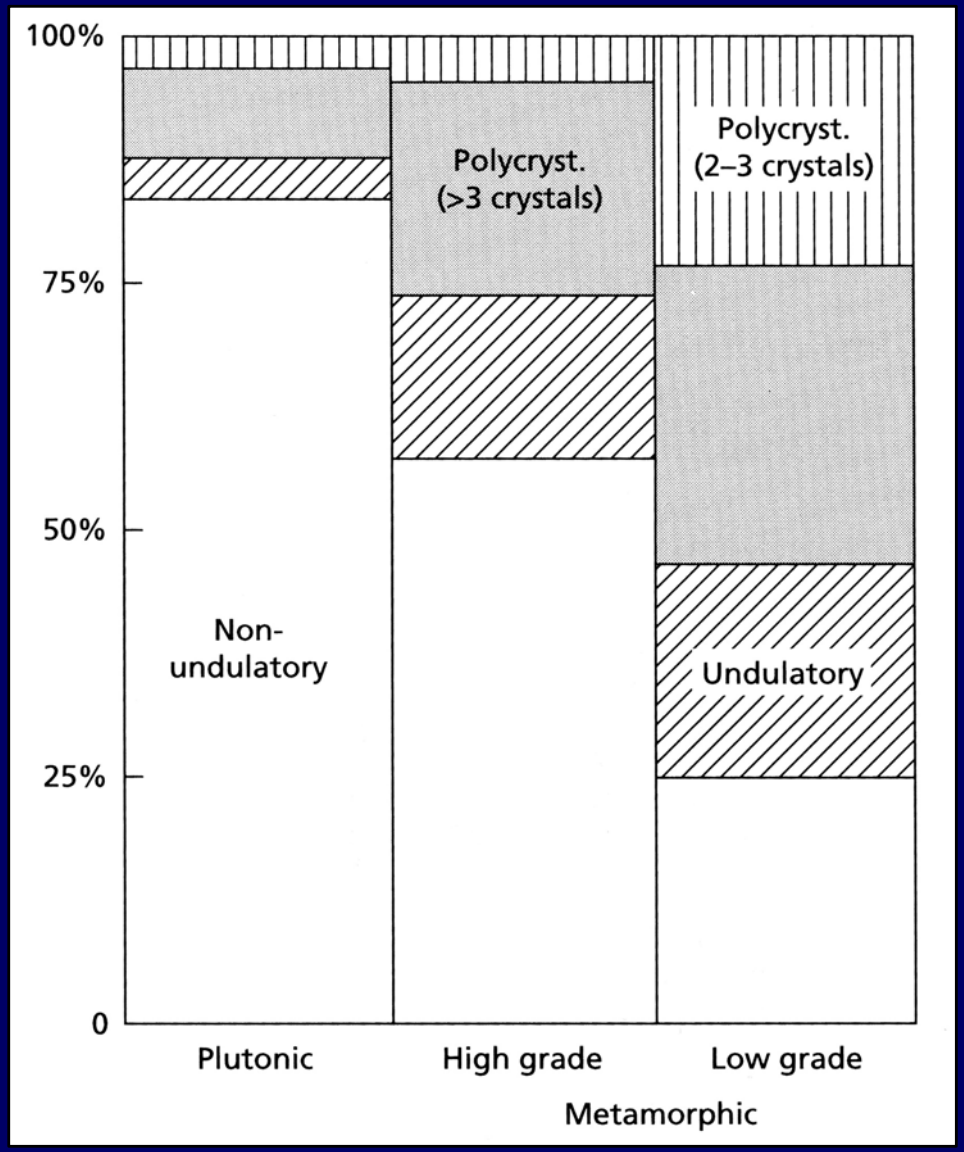
Quartz (SiO_2)

- Common source rocks have differences in the proportions of the various quartz types:

- Igneous rocks
  - Qm, non-undulatory, no inclusions

- Hydrothermal veins
  - Qm or coarsely Qp, but characteristically many fluid filled
  - vacuoles

- Low and high grade metamorphic and plutonic rocks might be distinguished by close examination of:
  - extinction properties
  - polycrystallinity



Relative abundance of detrital monocrystalline and polycrystalline quartz grains in Holocene sands derived from known plutonic and metamorphic sources (After Basu et al., 1975)

Feldspars ($KAlSi_3O_8$ – $NaAlSi_3O_8$ – $CaAl_2Si_2O_8$)

Second most important coarse detrital mineral in sedimentary rocks

Sharply subordinate to quartz in abundance

- av. sandstone has 10-12% feldspar
- feldspar content much more variable than quartz content (ranges between 0% to more feldspar than quartz)
- the average feldspar content of terrigenous rocks has increased from the early Paleozoic to the present



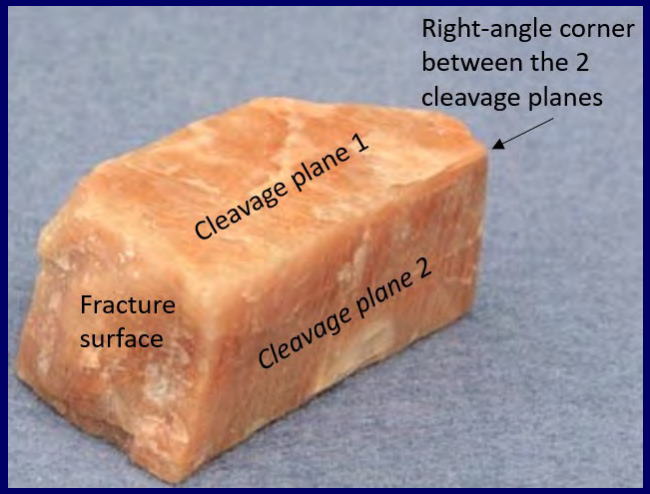
All varieties of feldspar are found in sedimentary rocks but:

- certain kinds of dominant, and others are rare
- Proportions of various feldspars very different in sedimentary rocks than in crystalline source rocks



Feldspars ($KAlSi_3O_8$ – $NaAlSi_3O_8$ – $CaAl_2Si_2O_8$)

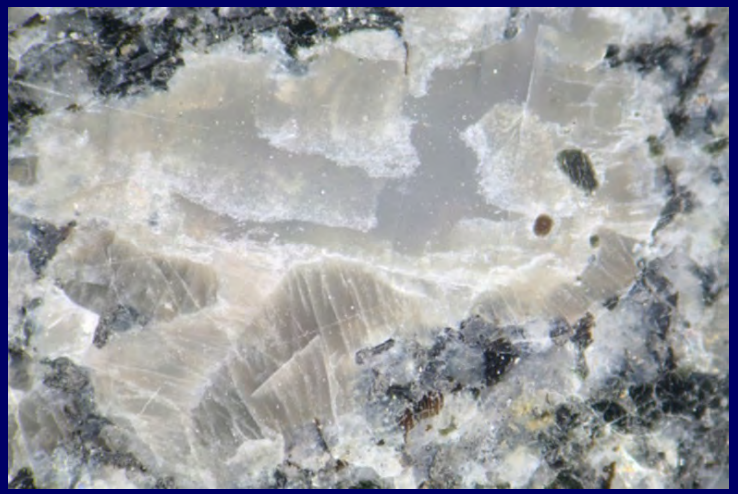
% of detrital feldspar in sedimentary rocks is significantly less than in the source rocks
only ~ 1/5 as much
Why?



Main characteristics of feldspar:
Mechanically weaker than quartz:

- cleavage
- softer

→ Abrade 2 x as fast as quartz but abrasion is still insignificant during transportation (at least in water)



Most important difference is that it is chemically weaker

- easily hydrolised

→ Unstable during weathering in soils and temporary storage during transportation

Chemical weathering of feldspars

Feldpars are metastable, and react readily with water and weak acids formed in water at surface T and P

$$H_2O + CO_2 \rightarrow H_2CO_3 \rightarrow H^+ + HCO_3^-$$

water carbon dioxide carbonic acid hydrogen ion bicarbonate ion

H+ can replace metal ion in some silicate minerals by a process called hydrolysis

Result:

- metal ion released into solution
- secondary minerals are formed

- Hydrolysis of K-Feldspar (orthoclase)

$$4KAlSi_3O_8 + 4H^+ + 2H_20 \rightarrow Al_4Si_4O_{10}(OH)_8 + 8SiO_2 + 4K^+$$

K-feldspar Hydrogen ion water Clay (Kaolinite) quartz K ion

- Hydrolysis of Plagioclase

$$2CaAl_2Si_2O_8 + 2H_2CO_3 + O_2 \rightarrow Al_4Si_4O_{10}(OH)_8 + 2CO_3^{2-} + 2Ca^{2+}$$

Ca-plagioclase carbonic acid oxygen Clay (Kaolinite) carbonate ion Ca ion

$$2NaAlSi_3O_8 + 9H_2O + 2H^+ \rightarrow Al_2Si_2O_5(OH)_4 + 4H_4SiO_4 + 2Na^+$$

Na-plagioclase water hydrogen ion Clay (Kaolinite) free ions Na ion

Chemical weathering of feldspars- Don't panic!!!

Feldpars are metastable, and react readily with water and weak acids formed in water at surface T and P

$$H_2O + CO_2 \rightarrow H_2CO_3 \rightarrow H^+ + HCO_3^-$$

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Na-plagioclase water hydrogen ion Clay (Kaolinite) free ions Na ion

Don't worry about learning the reactions for this Unit.

Chemical weathering of feldspars

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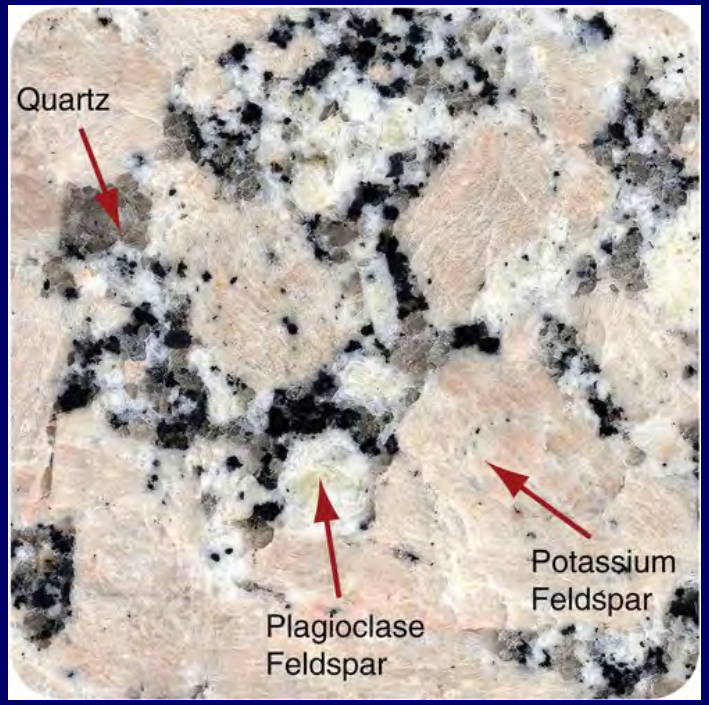
Ca-plagioclase carbonic acid oxygen Clay (Kaolinite) carbonate ion Ca ion

If you find a sedimentary rock with a lot of potentially unstable feldspar, like calcic plagioclase, there must have been some special conditions such that weathering did not have enough chance to operate (either not enough time or not enough intensity, or both)

Identification of feldspars in hand specimen

In hand specimen feldspars are white, cream, buff or pink, but never glassy. Sometimes powdery if strongly weathered.



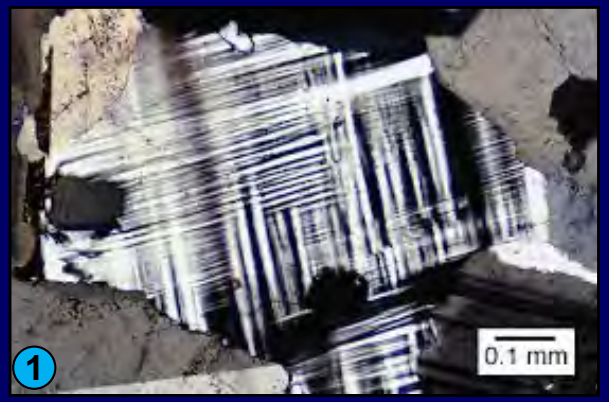


Identification of type of feldspar requires a thin section.

Identification of feldspars in thin-section

Potash feldspars (a.k.a. K-feldspar: orthoclase and microcline), are much more common in sandstones than plagioclases.

In thin-section:



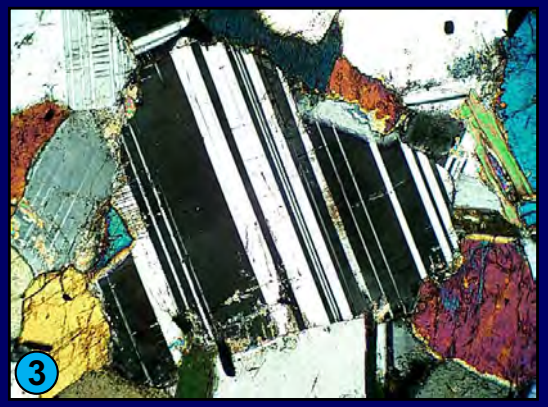
Microcline

- grid-iron twinning pattern
- somewhat more common that orthoclase but both common



Orthoclase

- crystals break along twin planes so Carlsbad twinning rarely seen
- low birefringence so often look like quartz
- look for alteration to distinguish



Plagioclase

- polysynthetic, multiple twinning
- albite (Na^+-rich) more common than anorthite (Ca^+-rich)
- comes from intermediate to mafic igneous rocks, as well as plagioclase-bearing metamorphic rocks (common in uplifted oceanic and island-arc terranes)

- common in continental basement rocks (coarse-grained granites and acid gneisses)

Lithic (‘rock’) fragments

Rock fragments are defined as particles with more than one mineral present

Dominate conglomerates and coarser particles in sandstones

As become smaller break into constituent minerals and grains

Not always easy to identify – fragments of fine-grained metamorphic rocks tend to be dark grey or black in hand specimen.



Can be of any rock, but metamorphic rocks are most common because many are particularly hard and durable

Micas



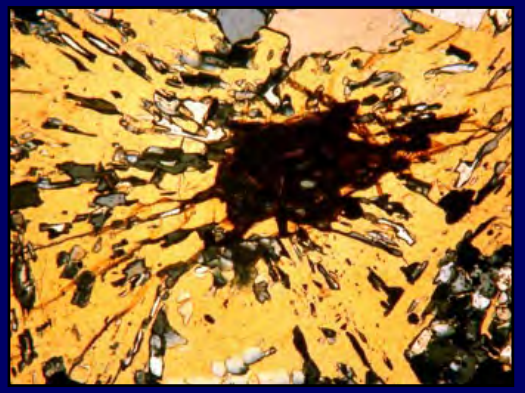
Detrital muscovite mica (and to a much lesser extent biotite) is common in many sedimentary rocks. Appears as flakes of silvery-looking mineral catching light on bedding surfaces. Derived from igneous rocks, metamorphic schists and phyllites.

Heavy minerals

- Accessory minerals <1% of sedimentary rocks but useful for identifying source rocks
- Predominantly silicates and oxides:
    - Common, non-opaque heavy mineral grains: apatite, epidote, garnet, rutile, staurolite, tourmaline and zircon
    - Common opaque detrital minerals: Ilmenite and magnetite
- Resistant to chemical and mechanical weathering

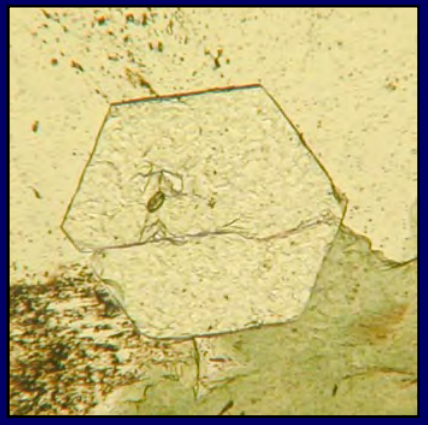


Euhedral zircon with growth zones, PPL (below), XPL (above)
Granite, NW Spain



Poikilitic porphyroblasts of staurolite, XPL





Perfectly euhedral basal section of apatite, PPL
Shap Granite, Cumbria



Garnet

All images: University College London

Other frequently occurring constituents

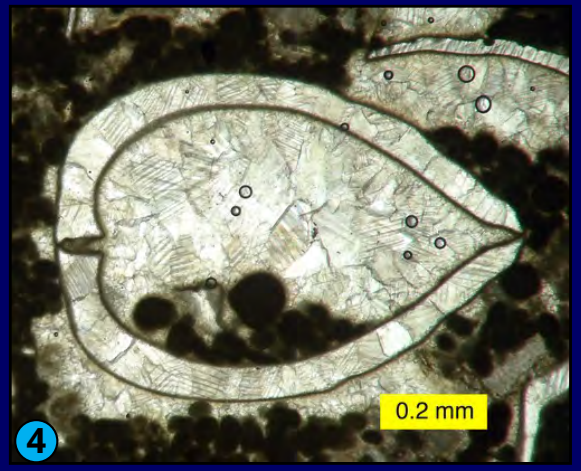




Calcium carbonate grains



Skeletal phosphate



Non-skeletal grains
(e.g., ooids, peloids etc)

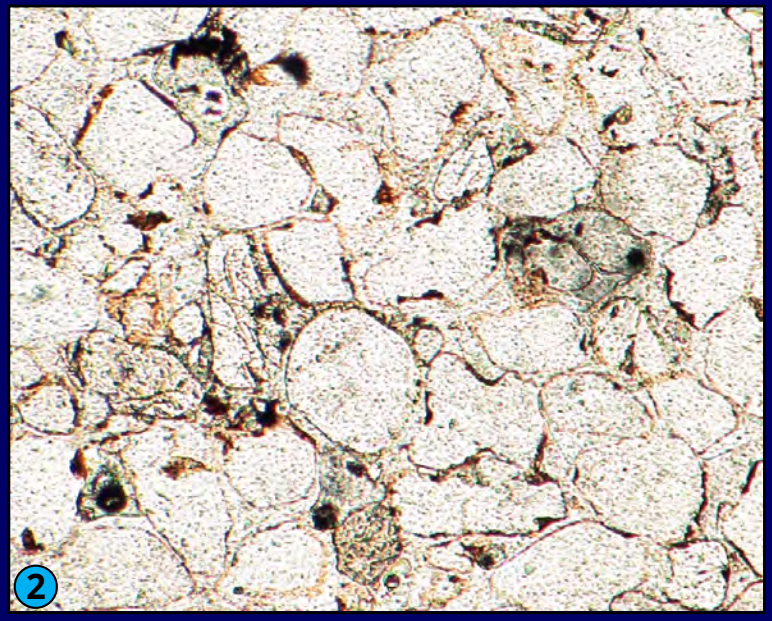




Carbonaceous plant fragments

Other frequently occurring constituents





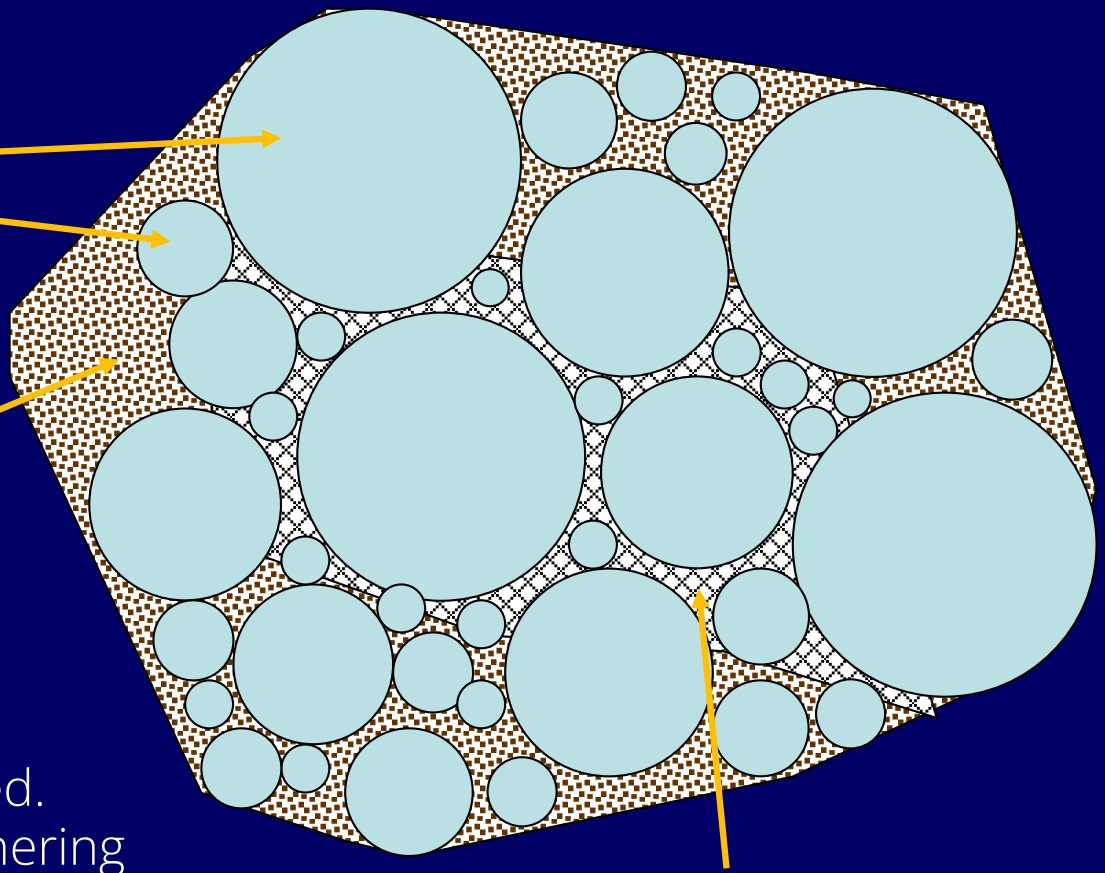
Moderately well rounded quartz grains with thin rims of hematite. PPL

Most sedimentary rocks contain some iron usually as oxides. These are responsible for yellow, red, brown and purple colours in many rocks:
Red = Hematite Fe_2O_3
Yellows, browns = Limonite/Goethite FeO.OH
Red sediments characteristic of rather arid terrestrial deposits

How to look at the geometry of a sediment or rock

In hand specimen or thin-section the standard way is:

Framework grains
a.k.a. clasts or particles
residual components of weathering i.e. quartz, feldspar and lithic fragments



Matrix
'fines'. Particulate material but altered. Typically products of chemical weathering such as clay minerals and silica

Cement
formed after deposition by precipitation into spaces

Matrix

Much is clay minerals produced from chemical weathering of feldspars and other aluminium-bearing silicates

Usually appears grey in hand specimen



Conglomerates: <2 mm

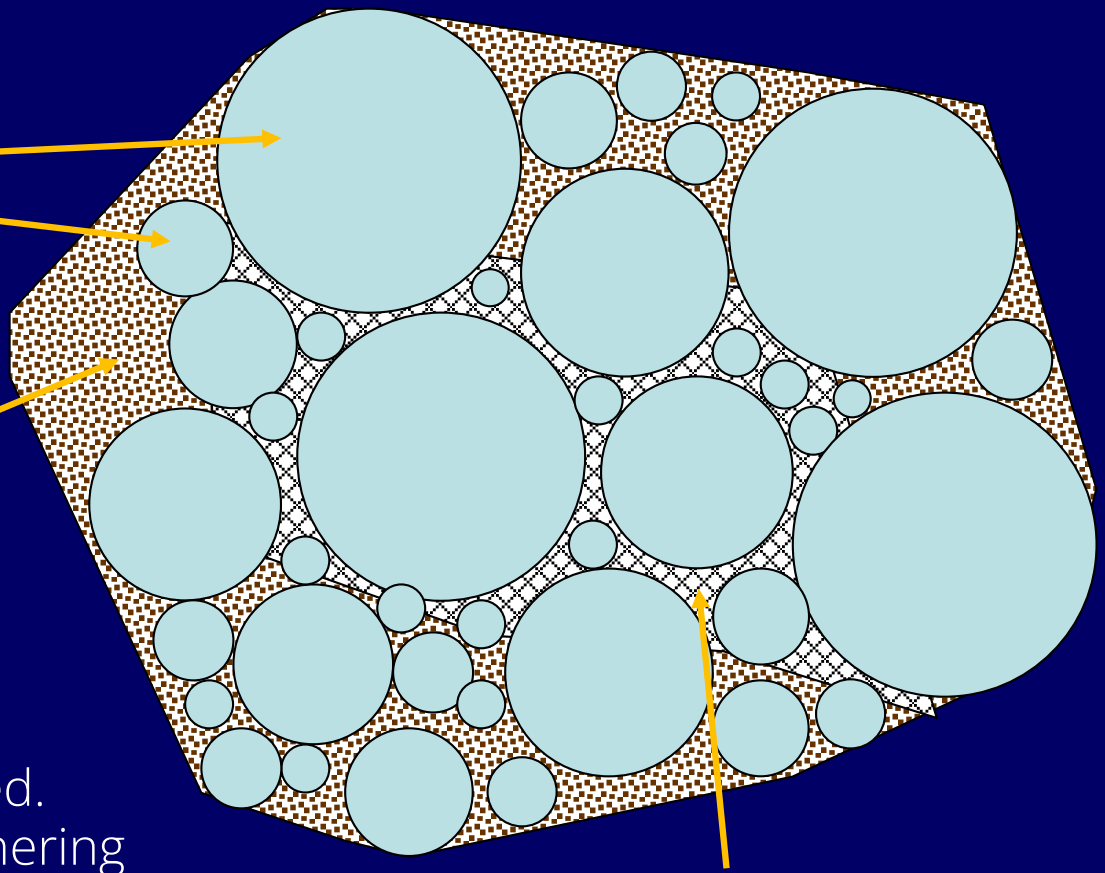


Sandstones: <30 μm widely used

How to look at the geometry of a sediment or rock

In hand specimen or thin-section the standard way is:

Framework grains
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Cement
formed after deposition by precipitation into spaces

Cements (form during diagenesis)

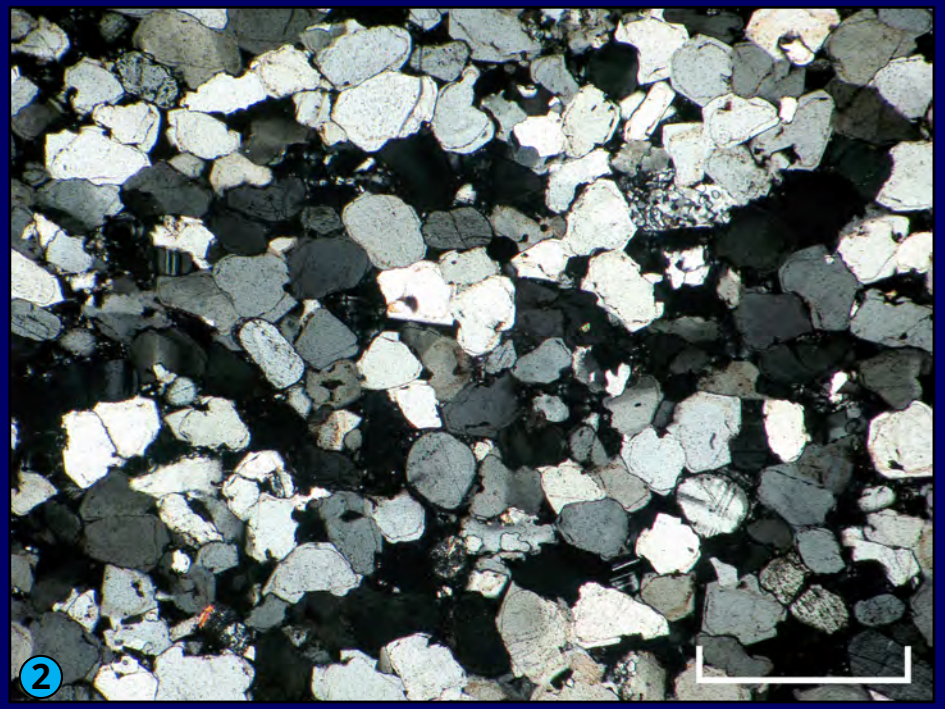
Precipitated into pore spaces *after* deposition.

Lithifying agent that turns sediment into sedimentary rock.

Only two common cements in sandstones:
1) Quartz – precipitated or authigenic quartz



Typical appearance of a quartz-cemented, quartz-rich sandstone when it becomes difficult to make out individual grains.



Siliceous cement and the secondary quartz overgrowths hold this rock together. Overgrowths of quartz in optical continuity with the original grain.

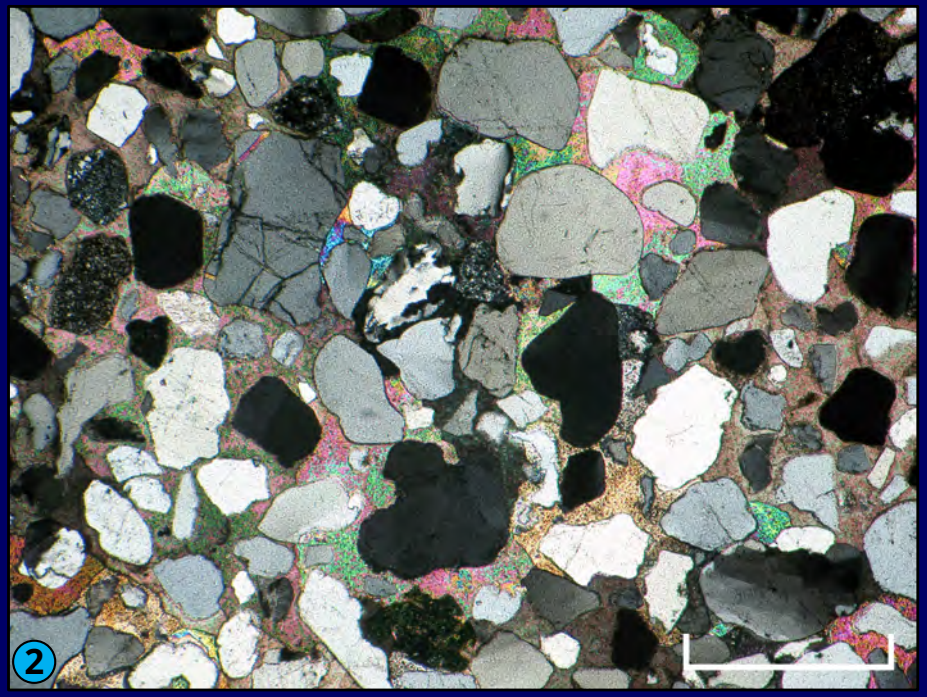
Cements

Only two common cements in sandstones:

2) Calcite – recognised by acid reaction



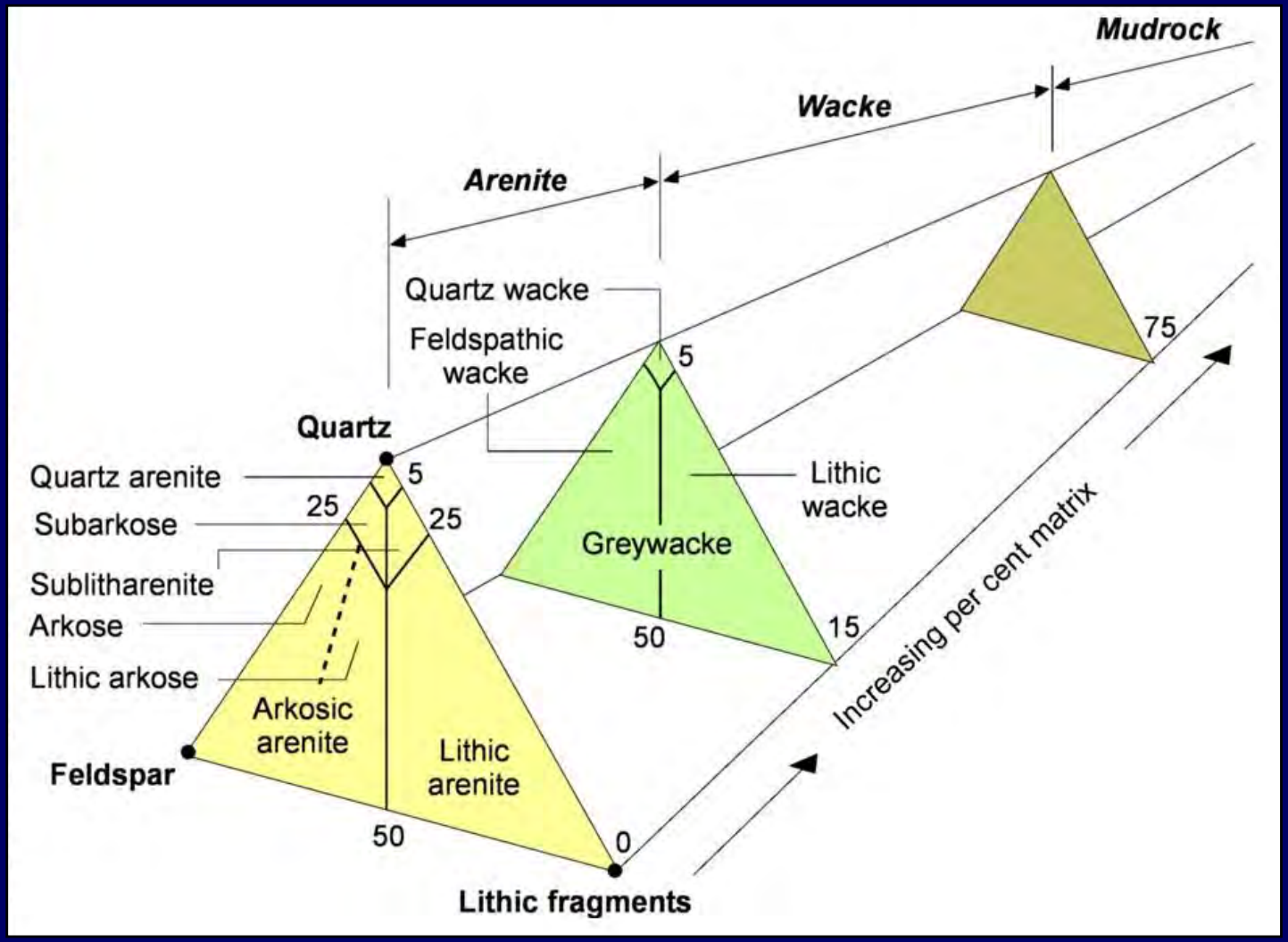
Mainly medium to coarse grained quartz sand. Poorly sorted. Small amounts of lithic fragments up to 10 mm across, and shell fragments.
Pale grey cement fizzes with acid, indicating that it is composed of calcium carbonate.



Quartz grains, sub-angular to sub-rounded, occasionally rounded, displaying undulose extinction. Lithic fragments including flint and quartzite. Some grains of feldspar. Grain supported. Cemented by a sparry calcite cement. The cement crystals are large, each enclosing a number of sedimentary grains.

Classification of Sandstones

Classification proposed by Dott and others is based on proportions of quartz, feldspar, rock fragments and matrix



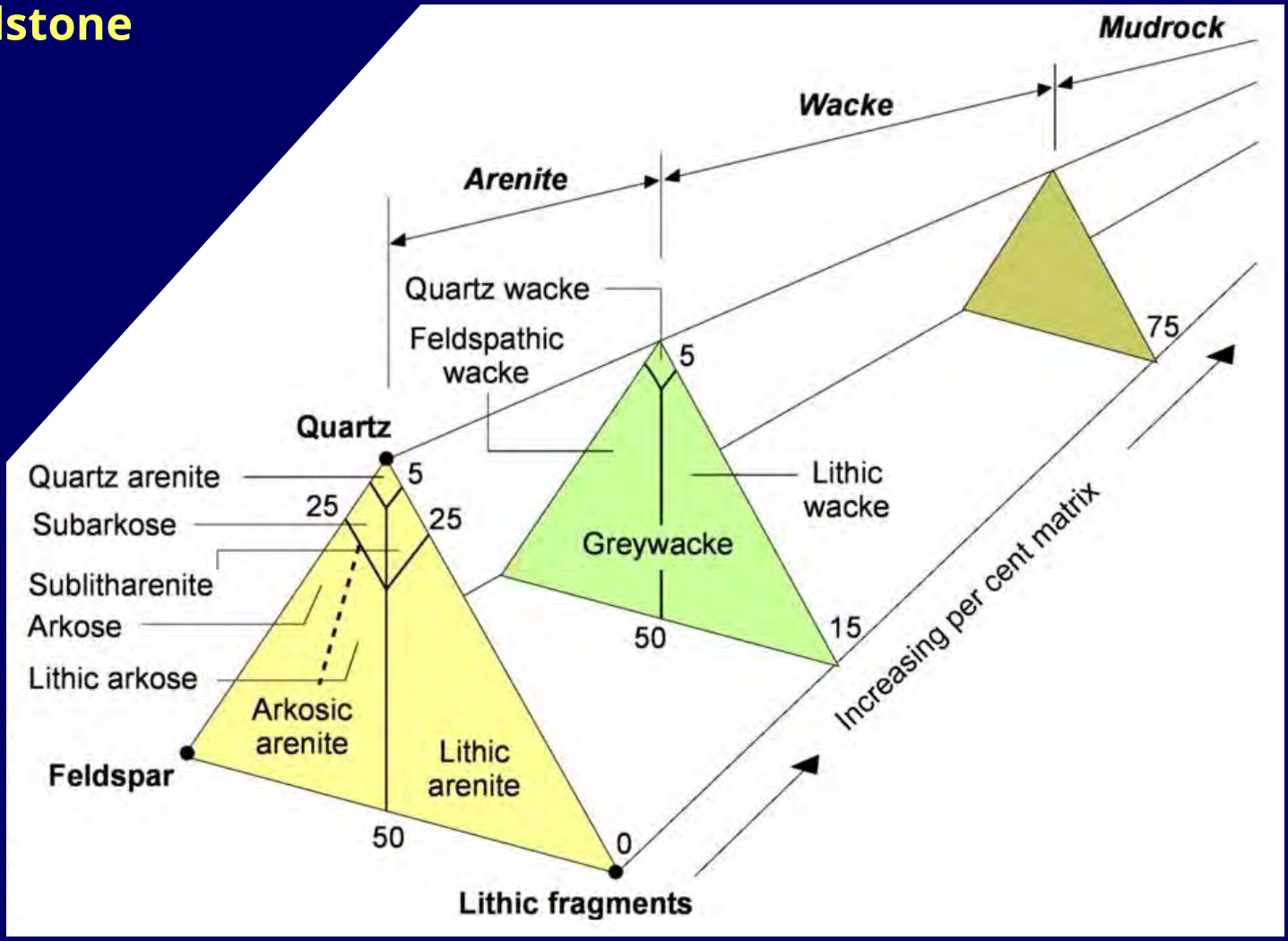
Classification of Sandstones

Firstly consider proportion of **matrix**:

<15% matrix = **Arenite**

15-75% matrix = **Wacke**

>75% matrix = **Mudstone**



Classification of Sandstones



Jurassic Estuarine Series,
North Yorkshire

Silurian Austwick Grit
Austwick, Yorkshire

Cambrian
Solva, South Wales

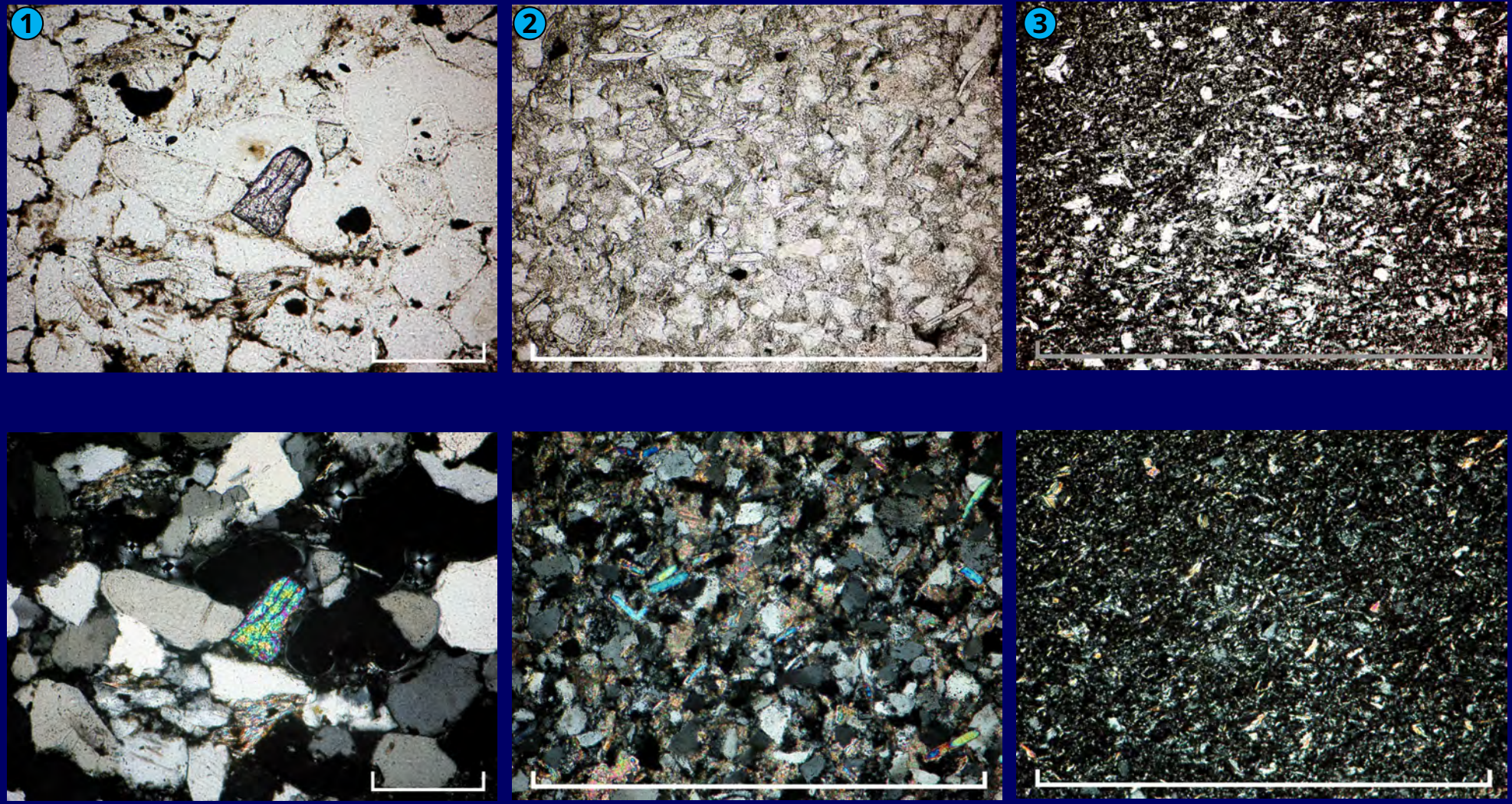
<15% matrix = Arenite

15-75% matrix = Wacke

>75% matrix = Mudstone

Scale bar: 1cm

Classification of Sandstones



<15% matrix = Arenite

15-75% matrix = Wacke

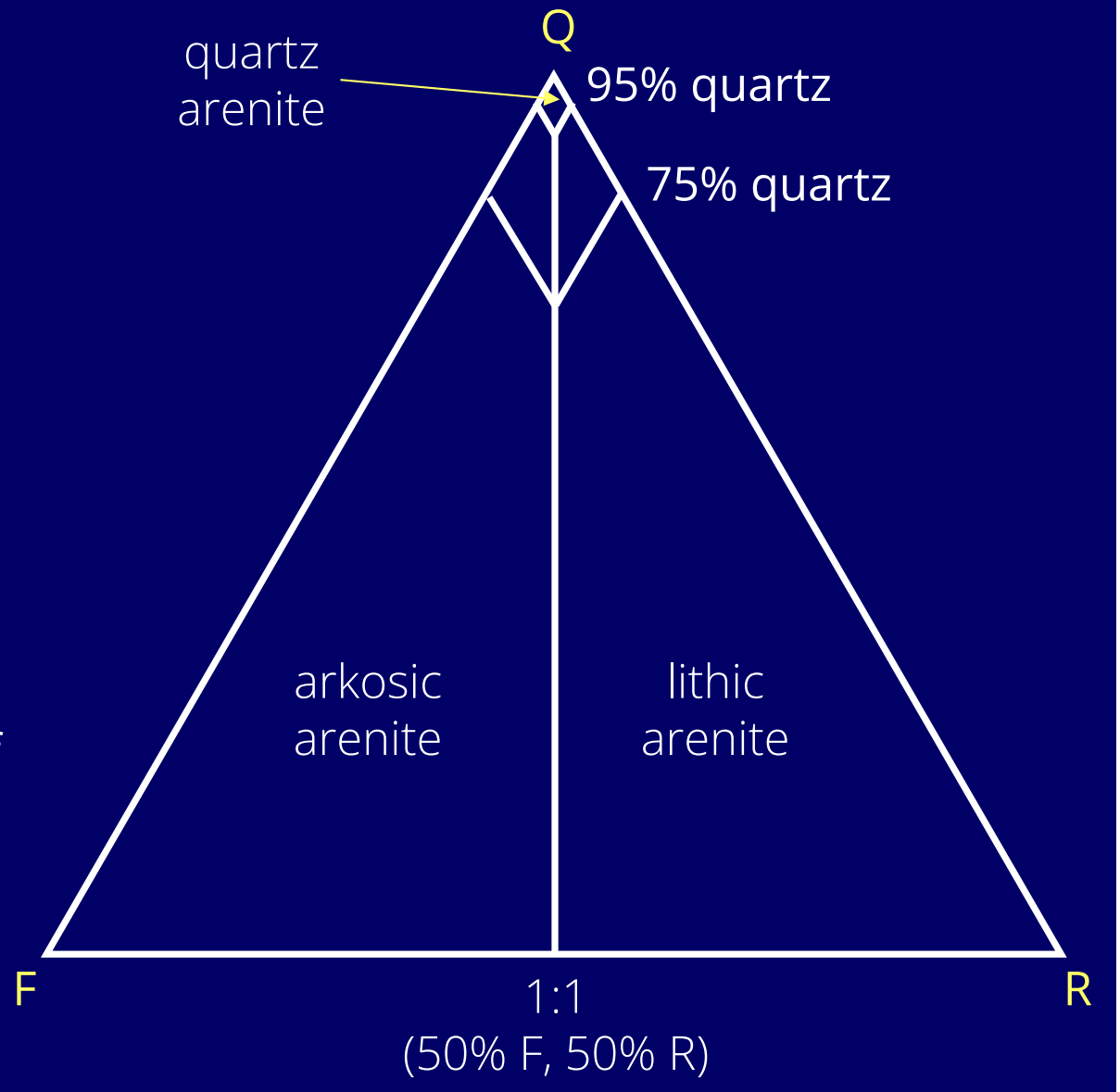
>75% matrix = Mudstone

Scale bar: 1mm

Classification of Sandstones

Secondly consider proportion of quartz (Q), feldspar (F) or lithic grains (R). For example:

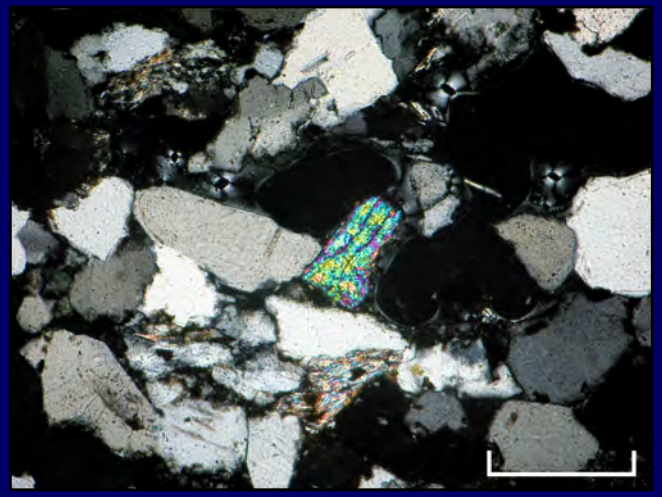
"Quartz arenite": arenite containing >95% quartz grains
"Arkosic arenite": arenite containing >25% feldspar and in which the amount of feldspar is greater than that of rock fragments
"Litharenite": arenite containing >25% rock fragments and in which the amount of rock fragments is greater than that of feldspar

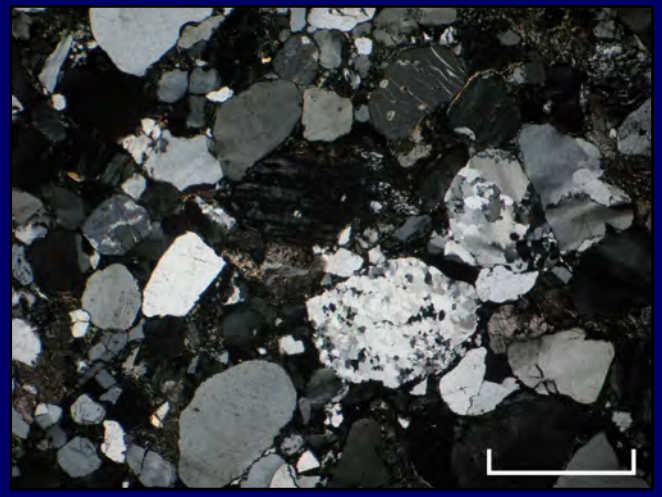


Classification of Sandstones





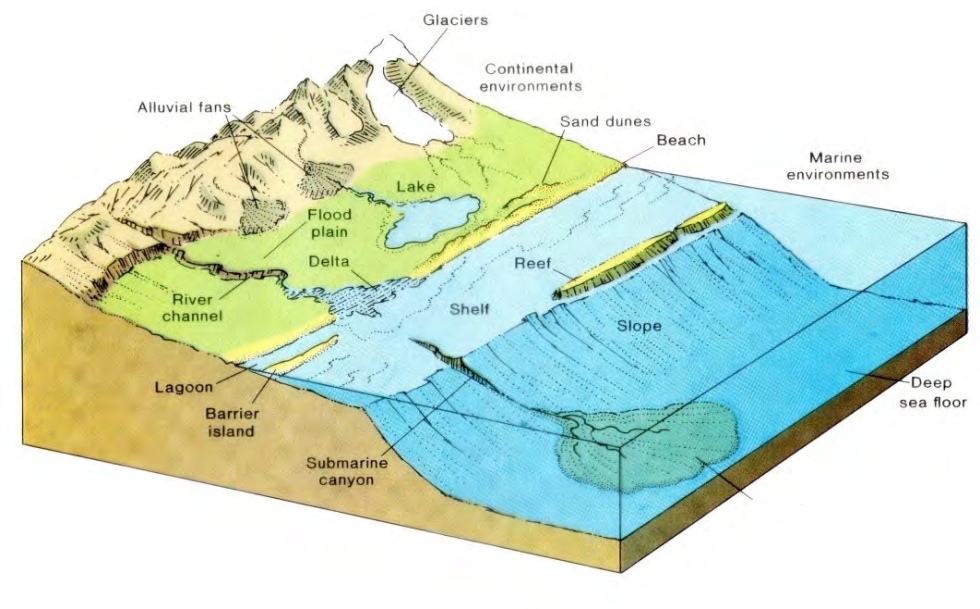




<15% matrix = Arenite
>95% Quartz = Quartz arenite

<15% matrix = Arenite
>25% feldspar = Arkosic arenite

Transportation on sediment



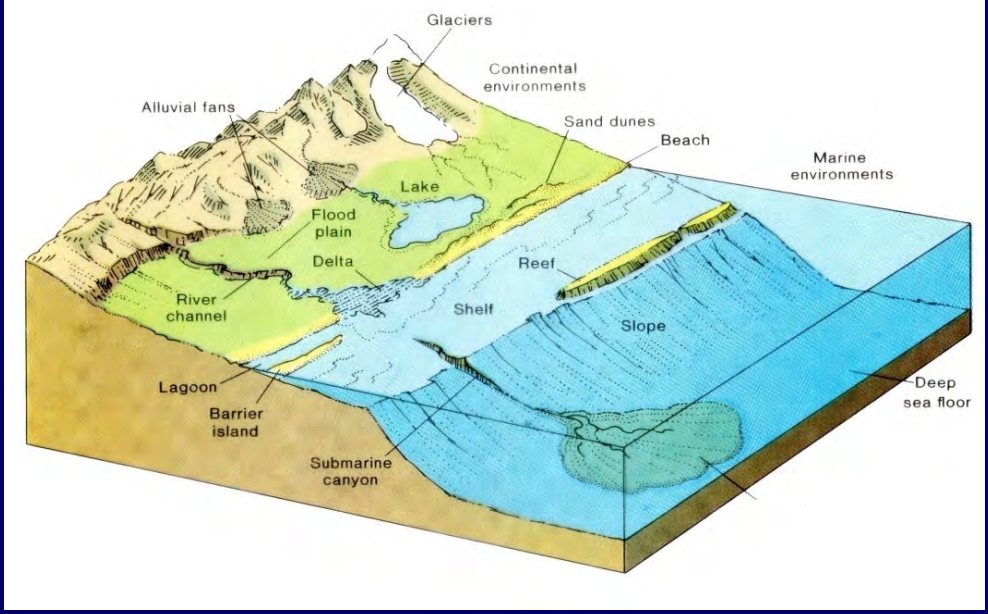
Weathered regolith and soil is eroded via fluvial (river), wind and glacial processes, and can be transported 10s to 1000s kms to a variety of environmental settings



Much fine-grained sediment is deposited at river mouths like here in the Ganges delta, India and Bangladesh

- 40,000 km^3 water runoff Earth's continents each year
- One volume of ocean water recycled via the rivers every 40k yrs

Transportation on sediment



Weathered regolith and soil is eroded via fluvial (river), wind and glacial processes, and can be transported 10s to 1000s kms to a variety of environmental settings



Weathering and erosion don't stop with removal of particles from source rock

- grains undergo abrasion as a result of grain to grain impacts during transport
- insitu weathering during temporary storage

Effect of transport on textural and compositional maturity

Sediment texture and composition all reflect the maturity and history of a sediment

Most changes are related to distance and time of transport and number of passes through the rock cycle

For example, in rivers changes in grain size, sorting, rounding and sphericity with transport are well documented

With increasing distance and time of physical transportation:

Grain size decreases





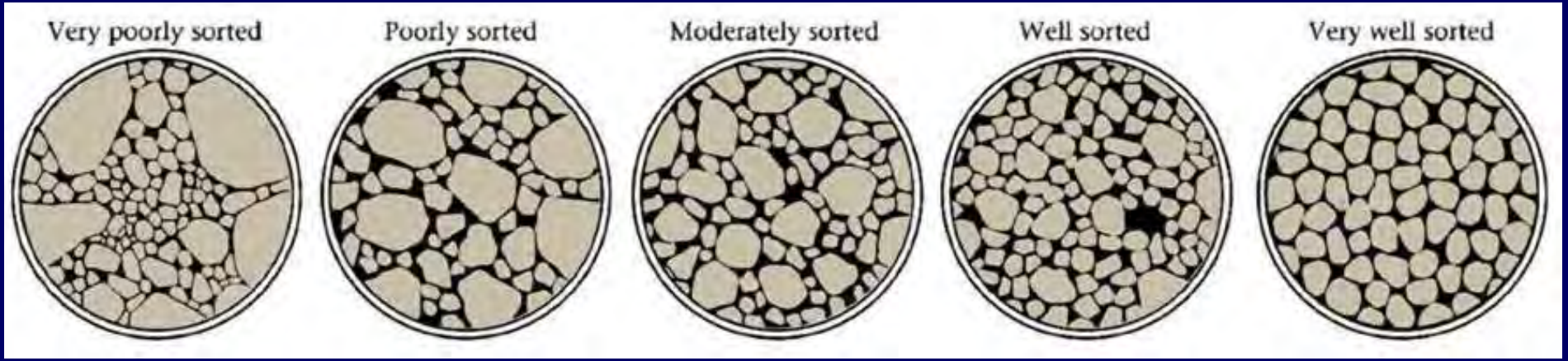


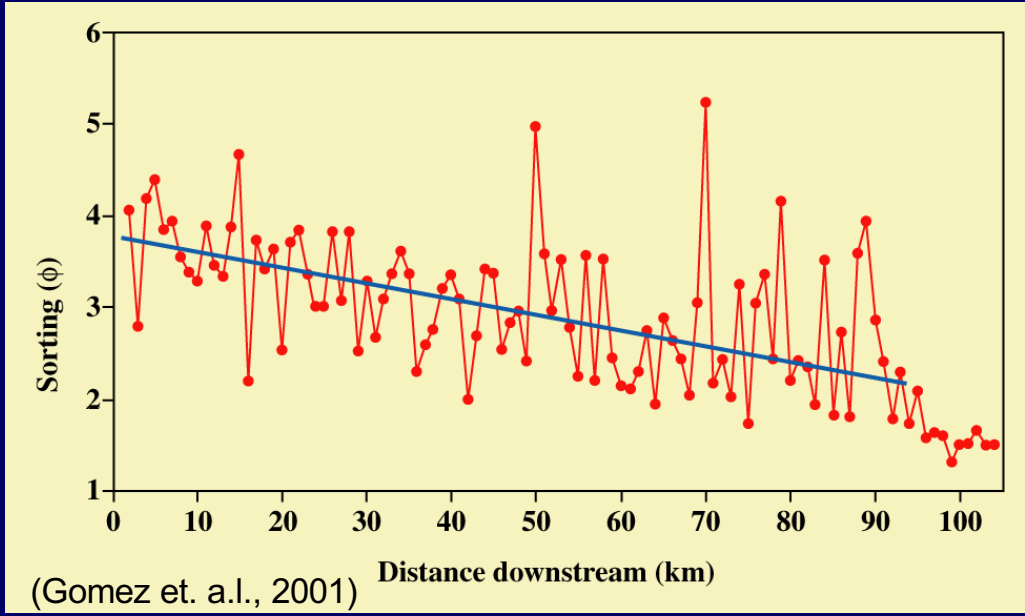
We will look at physical transport mechanisms later in the course

Effect of transport on textural maturity

With increasing distance and time of physical transportation:

Grain size sorting increases





(Gomez et. a.l., 2001)

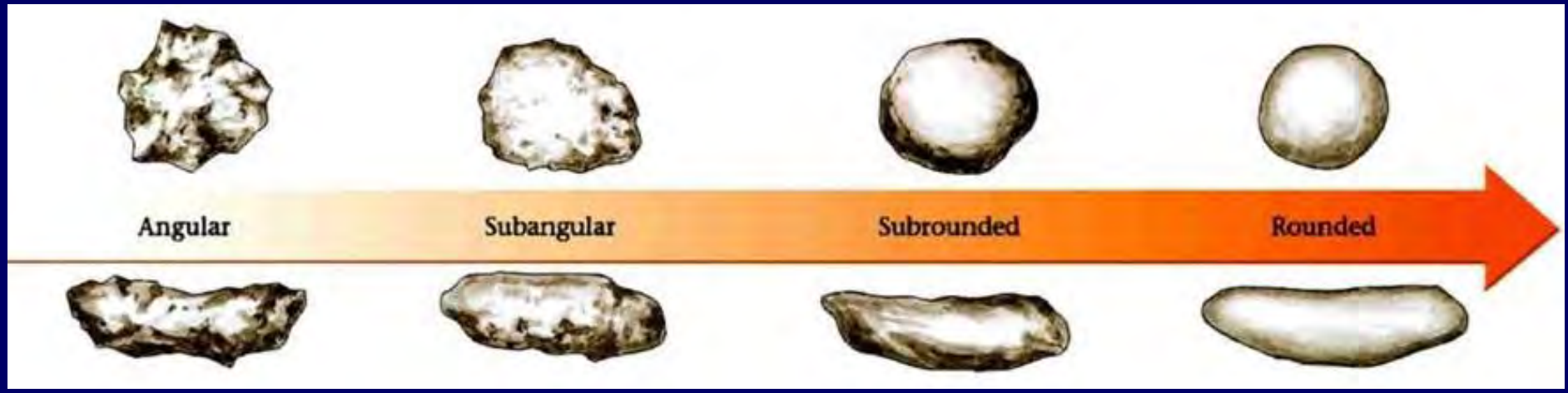
The longer the distance of transport the greater the opportunity to remove fine material (in suspension) and to leave coarser material behind, reducing the range of grain sizes present (improving the sorting and decreasing the sorting coefficient).

Effect of transport on textural maturity

With increasing distance and time of physical transportation:

Angularity decreases/ roundness increases rapidly

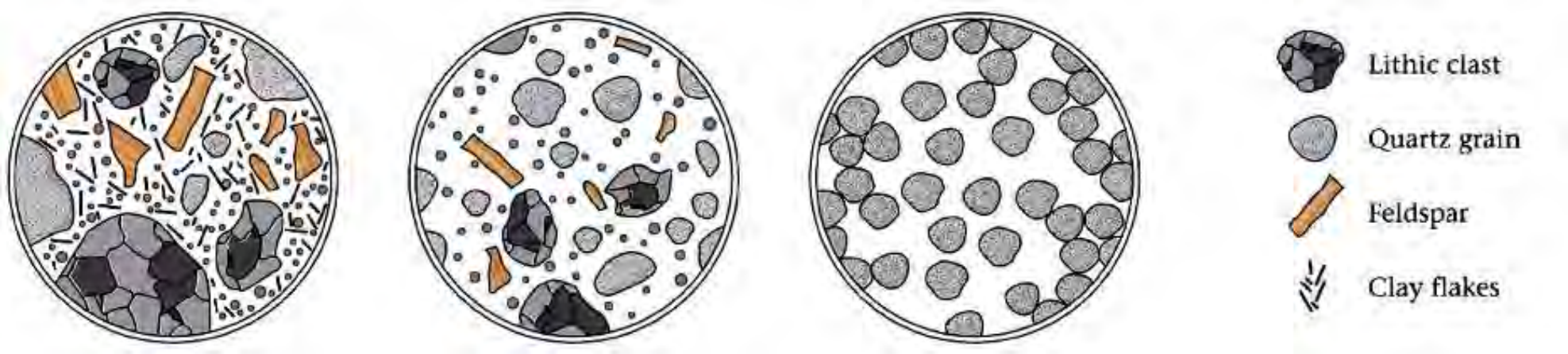
Sphericity increases slowly



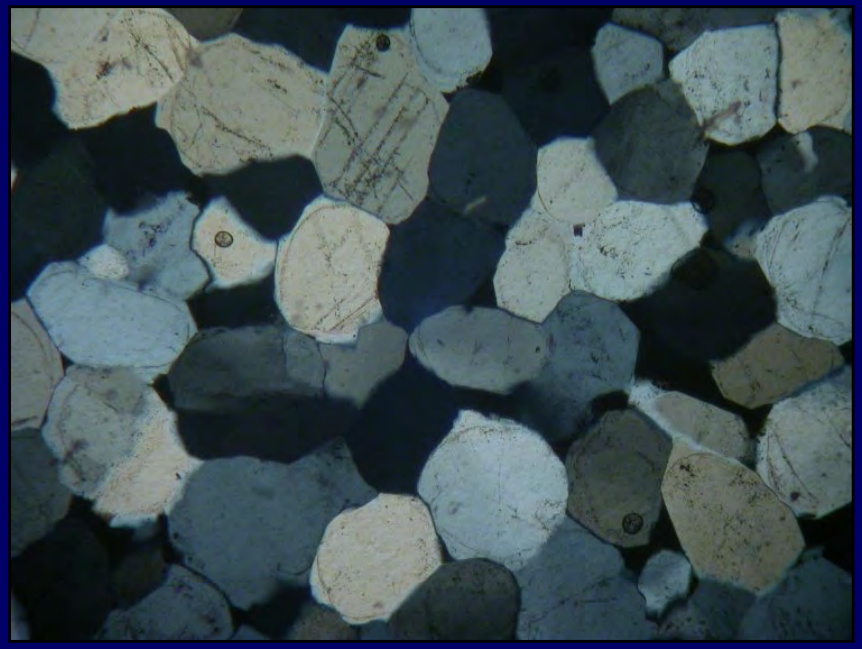
Effect of transport on compositional maturity

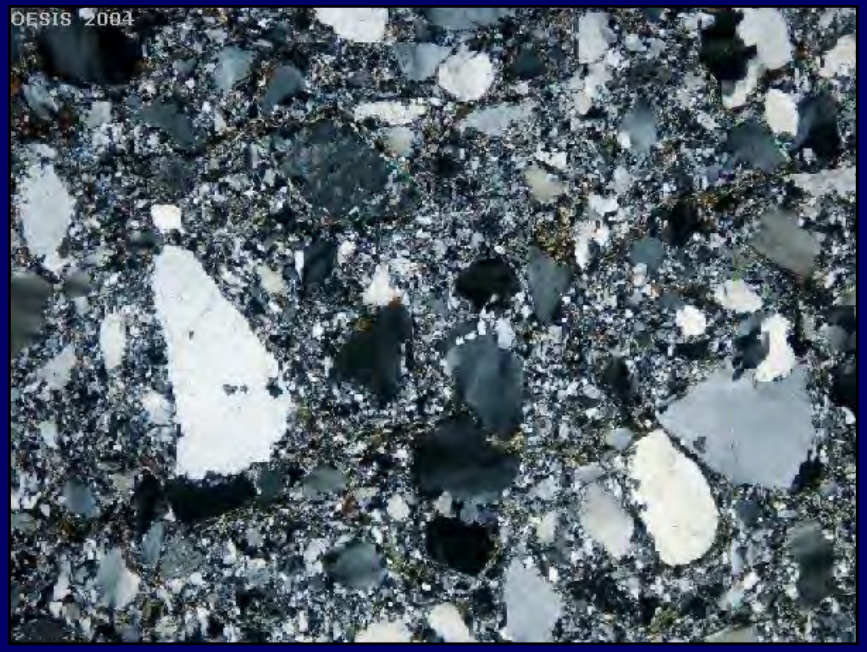
With increasing distance and time of physical transportation:

There is loss of feldspar, mica and other clasts in favour of clay matrix and quartz



The name of the sandstone tells you something of its maturity





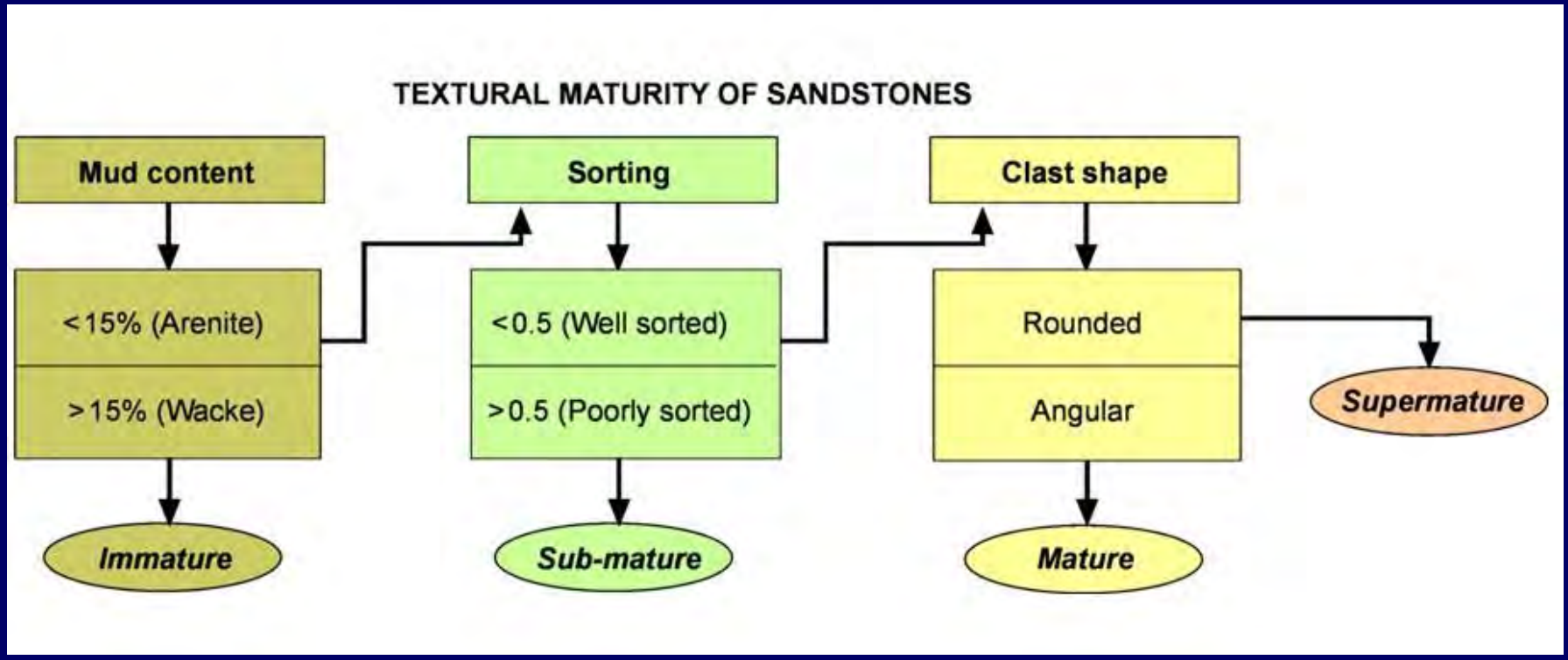
Quartz arenite <15% matrix

Quartz wacke >15% matrix

- better sorted
- more mature

(greater transport distance and/or more times through the rock cycle)

The name of the sandstone tells you something of its maturity



Practical Exercise 2

Complete Part B and Part C

What you should be able to do on conclusion of the class

1. make observations to describe the textural and compositional characteristics of a siliciclastic sediment or rock in hand specimen and thin section using appropriate nomenclature
2. graph grain size data and calculate standard statistics using the graphical method
3. explain what aspect of a grain size distribution each statistic describes
4. explain the role of weathering in determining the composition of siliciclastic rocks
5. recall and recognise in hand specimen and thin-section the main minerals which make up grains and cement in sandstones, and explain why matrix is predominantly made of clays
6. apply Wentworth's and Pettijohn's classifications to name a clastic sedimentary rock, based on your own observations
7. recall the impact on grainsize, texture, and mineralogy of sediment transport time (T) and distance (D)